Authoritarian Control through the Judiciary*

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Authoritarian governments may use courts to punish groups perceived as political threats. We examine this using 6.4 million Chinese criminal verdicts. Ethnic minorities receive 6.8% longer sentences and face 6.5% and 14.0% higher probabilities of life imprisonment and the death penalty, respectively. These additional penalties are particularly severe for crimes associated with social unrest, for less assimilated ethnic groups, and in regions experiencing heightened social tensions. Penalties against minorities intensified in minority-dominant prefectures during Uyghur unrest and when local political leaders sought promotions. The disparities, however, declined over time and almost disappeared in 2021; we discuss the potential causes.

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1 Introduction

All authoritarian regimes must maintain social order to ensure their survival. To prevent political opponents from challenging their authority, these regimes employ a variety of tactics—ranging from conventional methods based on violence and ideological indoctrination to surveillance, to information manipulation, to censorship, to local elections, and to economic policies (Andirin et al., 2022).[1] This paper provides empirical evidence on political control through the judiciary—a function of law in authoritarian regimes that has long been discussed but has received limited empirical examination (Solomon, 1996; Pereira, 2005). We argue that authoritarian governments can use criminal law to imprison individuals or groups perceived as potential sources of unrest and, in more extreme cases, impose life imprisonment or the death penalty.

Our empirical analysis examines the treatment of ethnic-minority defendants in China's criminal justice system. Han Chinese account for approximately 91% of the population, while China officially recognizes 55 ethnic-minority groups with diverse languages, religions, customs, and cultural traditions. Because the Chinese Communist Party (CCP) faces little organized political opposition, ethnic unrest can pose a particularly salient threat to state control. The Party controls the judiciary; about 85% of judges and virtually all court presidents are Party members. This structure may lead to unfavorable treatment of ethnic-minority defendants perceived as threats to the central government.

This paper assembles, to our knowledge, the most comprehensive dataset on China's criminal justice system. Our baseline sentence-length analysis uses 6.4 million observations from publicly available criminal court records covering 2013 to 2021. Because the Supreme People's Court required local courts, in principle, to publish all criminal verdicts, this dataset covers nearly the universe of criminal sentencing in China. Using text-processing and machine-learning methods, we extract an extensive set of detailed case and defendant characteristics from unstructured judicial documents. These data allow us to compare sentencing outcomes between ethnic-minority and Han defendants across criminal offenses nationwide.

Drawing on these comprehensive data and comparing observably similar Han and ethnic-minority defendants, we find that minority defendants receive 6.8% longer sentences and are 6.5% and 14.0% more likely to receive life and death sentences, respectively. For comparison, the estimated disparities are of the same order of magnitude as the Black–White sentencing

[1]For reviews, see Gehlbach et al. (2016), Guriev and Treisman (2019), and Egorov and Sonin (2024). For specific tactics, see Olson (1993) on violence; Adena et al. (2015) and Yanagizawa-Drott (2014) on ideological indoctrination; Xu (2021) and Beraja et al. (2023) on surveillance; Rozenas and Stukal (2019) on information manipulation; King et al. (2013) and Chen and Yang (2019) on censorship; Martinez-Bravo et al. (2022) on local elections; and Wen (2024) on economic policies.

disparities documented in the United States (Anwar et al., 2012; Rehavi and Starr, 2014; Cohen and Yang, 2019; Feigenberg and Miller, 2021).

The results also reveal substantial heterogeneity. The minority penalty is larger for drug, violent, and sex offenses, which are often viewed as particularly serious or associated with organized activity. It is also larger for minority groups associated with Islam, Christianity, and Tibetan Buddhism and for ethnic groups with a history of conflict with the Han majority. These groups tend to be less assimilated into mainstream Chinese society. The additional penalty is also greater in poorer, high-minority-share, and high-crime prefectures, where the potential risk of social disorder may be greater. Han judges, who preside over about 90% of criminal cases, also impose larger minority penalties than minority judges.

To examine the judiciary's role in maintaining political control, we ask whether courts punish ethnic minorities more harshly when threats of unrest increase (Costa and Zhao, 2025). Uyghur separatist movements in Xinjiang provide a unique setting to study such possibilities, as they may fuel fears that other minority groups will mobilize elsewhere in China (Wen, 2024). We find that during months with Uyghur conflict events, the minority–Han sentencing gap widens significantly more in high-minority-share prefectures than in low-minority-share prefectures. The increase in sentence length is about 24% of the baseline gap. The increases in the probabilities of life imprisonment and the death penalty are about 37% and 44% of the respective baseline gaps in the full sample. The increase in sentence length is concentrated among working-age men, whom officials may view as the main participants in violent unrest. In contrast, we find little change in sentence length among women or older men.

We further investigate how local officials' career incentives shape sentencing decisions. In China's bureaucracy, prefectural leaders have strong incentives to maintain social stability because major incidents such as protests or riots may preclude promotion and, in some cases, effectively end an official's political career. We focus on China's distinctive age-based promotion system, under which officials generally lose eligibility for promotion after age 55. We find that the minority–Han sentencing gap widens more under promotion-motivated leaders. The increase in the sentence-length gap is equivalent to 8% of the baseline minority penalty.

Our findings indicate that China's judiciary plays an active role in maintaining state control by punishing ethnic-minority defendants more harshly than observably similar Han defendants. Unfavorable treatment of ethnic minorities may have intensified over time, as the international community has raised concerns about China's harsher policies toward minorities, especially since 2017 (OHCHR, 2022). In stark contrast, our results show that minority–Han sentencing disparities decline substantially over time, with the gap in sentence length disappearing in 2021. In the conclusion, we discuss potential causes, particularly how

nationwide legal reforms to increase judicial transparency and standardization may have affected ethnic sentencing disparities.

2 Background

2.1 China's Ethnic Minority Issues

Ethnic minority issues lie at the heart of social stability in China. Because the Chinese Communist Party (CCP) remains the sole ruling party and faces virtually no organized political opposition, it regards the governance of minority communities as central to maintaining political and social stability.

Ethnicity has long been a source of political conflict in China. China's ethnic policies draw heavily on the Soviet model, granting minorities limited autonomy while seeking to prevent territorial secession. Since the founding of the People's Republic of China, the central government has formally recognized rights to regional ethnic autonomy, although these rights have rarely been fully implemented. Religious freedom, cultural and linguistic preservation, and equal protection under the law have not always been guaranteed, contributing to disputes between the Chinese government and ethnic minorities (Zhang, 2011). These tensions have become increasingly visible, with unrest in Tibet and Xinjiang in 2008 and harsher policies toward Muslim minorities since 2017.[2] These policies raise concerns about unjust imprisonment or disproportionately harsh sentences for ethnic minorities.

At the same time, the Chinese government has implemented various preferential policies benefiting ethnic minorities (Sautman, 1998). These policies have included one-child policy exemptions, extra points on national college entrance exams, tuition waivers, stipends, job preferences, and tax benefits. Minorities also have the right, at least formally, to preserve their languages and cultures. Targeted poverty-alleviation and local economic-development programs provide further support. Together, these policies could lead one to expect favorable treatment for ethnic minority defendants in the judicial system.

China's broader approach to ethnic governance helps explain the tension between preferential and coercive policies. The state seeks to build a "unitary multiethnic nation," maintaining political integration while recognizing ethnic diversity and supporting minority development (Lai, 2009). These competing objectives can produce inconsistent policies across institutions. Whether, and under what circumstances, courts treat ethnic minority defendants more leniently or more harshly than comparable Han defendants is therefore ultimately an empirical question.

[2]These policies include mass surveillance and large-scale detention and have prompted allegations of crimes against humanity (OHCHR, 2022; Al Jazeera, 2018).

2.2 China's Criminal Justice System: State Control and Existing Evidence

China's criminal justice system can function as an instrument of state control for two reasons (see Appendix A-1 for details on criminal procedure). First, courts operate under extensive political oversight and are expected to support Party objectives, including maintaining social order (Li, 2016; Ng and He, 2017). The Party influences judges' career advancement and discipline, and courts are explicitly expected to "pursue political effects" in adjudication. More than 85% of judges were Party members in 2016, as were virtually all court presidents and vice presidents (Zhao, 2016; Woo, 1999). Second, China follows the civil-law tradition, in which judges play a central role in determining facts, applying the law, and issuing rulings (Ng and He, 2017). Historically broad sentencing ranges and limited guidance give judges substantial discretion (Ash et al., 2026; Liu et al., 2026), creating a channel through which political pressure to maintain social order may lead to disproportionately severe sentences for ethnic minority defendants.

Despite widespread attention to China's criminal justice system, systematic empirical evidence on ethnic disparities remains limited. Existing quantitative studies focus on particular offenses or groups, and their findings are mixed. Some find more lenient treatment of ethnic minorities (Peng and Cheng, 2022; Tian, 2023), others find harsher treatment (Hou and Truex, 2022; Steinert, 2023), and others find no significant disparity (Li et al., 2018).

3 Data

3.1 A Near-Complete Database of Criminal Court Judgments

We assemble a comprehensive database of criminal sentencing judgments in China from China Judgments Online (CJO) (CJO, 2021). Since 2014, the Supreme People's Court has required Chinese courts to publish judgments online. Because courts sometimes delay publication and CJO sometimes deletes judgments, we automatically collect and archive CJO judgments daily.

We use text-processing and deep-learning methods to transform millions of unstructured and nonstandardized judgments into a structured dataset containing an exceptionally rich set of variables describing each case (Appendix B-1). The data cover 3,506 courts and 49,187 judges. We extract court and judge identifiers, sentencing dates, and defendants' age, gender, ethnicity, education, occupation, and criminal history. We also predict judges' ethnicity, classifying 9.3% as ethnic minorities (see Appendix D for details). Cases span 427 crime types, providing a much finer classification than those used in existing studies. We also include detailed aggravating and mitigating factors, such as offense severity, guilty pleas, and voluntary surrender (Table B1). Because courts convict more than 99.99% of

defendants, we measure sentence severity using the length of fixed-term imprisonment and indicators for life imprisonment and the death penalty. Approximately 0.31% and 0.16% of cases result in life imprisonment and the death penalty, respectively.

Then, we construct the main analytical sample in the following way (Appendix B-2). First, CJO does not publish juvenile cases or certain politically sensitive cases. Second, we focus on first-instance criminal judgments because more than 95% of second-instance proceedings leave the original judgment unchanged. Finally, we exclude judgments with missing sentence length or defendants' ethnicity. The final sentence-length estimation sample comprises 6,434,145 cases, representing 73.4% of the 8,769,103 records in the input dataset. The retained sentence-length sample is broadly similar to the input dataset in observed defendant and case characteristics (Table B2), partly alleviating concerns about sample selection. See Table B3 for summary statistics.

3.2 Uyghur Unrest and Prefectural Political Leaders

We manually construct a dataset of protests and violent conflict associated with Uyghur unrest in Xinjiang using reports from Wikipedia, Human Rights Watch, and BBC News (Wikipedia, nd; HRW, nd; BBC News, nd). The dataset contains 83 events occurring between 2013 and 2017, and we plot the monthly event counts in Panel B of Figure F1. Our sources report no events after 2017, when the government substantially intensified its repression in Xinjiang.

We compile biographical data on 3,030 prefectural mayors and Party secretaries serving between 2014 and 2021, using official résumés and government appointment and dismissal reports to identify leadership transitions. Following the age-based promotion system described in Section 6.1, we classify leaders younger than 55 when taking office as "promotion-seeking" and those aged 55 or older as "less promotion-motivated."

Prefectural minority population shares are measured using the 2010 census (Panel A of Figure F1). These shares define the high- and low-minority-share prefectures used in the leader analysis.

3.3 Unconditional Sentencing Disparities

Figure 1 presents time trends of unconditional disparities in sentence length (Panel A) and the probabilities of life imprisonment (Panel B) and the death penalty (Panel C) between Han and ethnic minority defendants, with bars showing the number of cases in each group. On average, ethnic minority defendants receive harsher penalties than Han defendants. These unconditional disparities appear to narrow rapidly over the study period.

4 Ethnic Minorities in China's Criminal Justice System

4.1 Measuring Conditional Ethnic Disparities in Criminal Sentencing

We measure ethnic disparities in sentencing following Rehavi and Starr (2014), Cohen and Yang (2019), and Feigenberg and Miller (2021). We do not directly adopt existing methods based on the quasi-random assignment of cases to judges for two reasons. First, conventional judge-leniency designs often estimate the effects of judges' decisions, such as imprisonment (Bhuller et al., 2020). Our interest, however, is in sentencing differences by ethnicity rather than the effects of these decisions. Second, Arnold et al. (2022) assess racial discrimination in bail decisions using whether released defendants miss court or are rearrested.[3] However, we have no comparable outcome measure for sentencing, as whether a sentence is appropriate is essentially a normative question.

We compare cases within the same crime categories and jurisdictions and before the same judges, while controlling for detailed case circumstances:

$$Sentence_{icpjt} = \alpha + \beta Minor_i + \boldsymbol{X}'_{icpjt}\theta + \lambda_c + \gamma_p + \sigma_j + \delta_t + \epsilon_{icpjt}. \quad (1)$$

The dependent variable, $Sentence_{icpjt}$, is either sentence length in months, an indicator for life imprisonment, or an indicator for the death penalty. Defendant i's case is decided by judge j in court c, located in province p, during year-month t. Because life imprisonment and the death penalty can be imposed only for severe offenses, we estimate the corresponding specifications using the relevant subsamples. $Minor_i$ equals one for an ethnic minority defendant and zero for a Han defendant. The specification includes court fixed effects (λ_c), province fixed effects (γ_p), judge fixed effects (σ_j), and year-month fixed effects (δ_t). These fixed effects account for differences in legal practices across jurisdictions, nationwide changes in laws and legal practices over time, and time-invariant differences across judges.

The vector $\boldsymbol{X}_{icpjt}$ contains an extensive set of defendant and case characteristics described in Section 3, including indicators for 427 crime types and the aggravating and mitigating factors collected from the judgments. We also interact relevant case indicators with quantitative measures. For example, we interact an indicator for heroin possession with the weight of the drugs involved. These controls allow us to compare Han and ethnic minority defendants within narrowly defined crime types and under similar offense circumstances.

The coefficient of interest, β, captures the average difference in sentencing outcomes between ethnic minority and Han defendants, conditional on the included covariates and fixed effects. Standard errors are clustered at the court level.

[3]The model assumes judges seek to prevent pretrial misconduct, which is observed only among released defendants.

4.2 Five Facts about Ethnic Minorities in China's Judiciary

4.2.1 Fact 1: Harsher penalties for ethnic minorities

Ethnic minority defendants receive significantly harsher penalties than observably similar Han defendants. Figure 2 shows that minority defendants receive 6.8% longer sentences than observably similar Han defendants and are 6.5% and 14.0% more likely to receive life imprisonment and the death penalty, respectively (1. Baseline).[4] These differences are economically meaningful. Eliminating the disparity would have reduced aggregate prison terms for ethnic minority defendants by approximately 77,499 years and the numbers of life and death sentences by 256 and 197, respectively, during our study period.[5] For comparison, U.S. studies find that Black defendants receive sentences 10–20% longer than non-Black defendants (Anwar et al., 2012; Rehavi and Starr, 2014; Cohen and Yang, 2019; Feigenberg and Miller, 2021). Panel A of Table E1 reports the estimates.

In China, differences in crime and case characteristics explain a large share of the unconditional sentencing disparity between Han and ethnic minority defendants. In Table E2, we sequentially add controls across columns (1)–(6). Without controls, minority defendants receive sentences that are 2.90 months (17.2%) longer than those of Han defendants (column (1)). Adding crime-type and case controls reduces the unconditional sentence-length gap by about 75%, to 0.71 months (column (2)). The remaining controls, including province, court, and judge fixed effects, produce comparatively smaller changes. This pattern is consistent with U.S. evidence that offense and case characteristics explain much of the unconditional Black–White sentencing gap (Rehavi and Starr, 2014). For life imprisonment and the death penalty, however, province fixed effects play a larger role in reducing the disparities (Panels B and C).

4.2.2 Fact 2: Larger minority penalties for severe and organized crimes

The "minority penalty"—the additional penalty imposed on ethnic minority defendants relative to observably similar Han defendants—is more pronounced for violent crimes (2 in Figure 2), drug-related offenses (3), and sex crimes (4), which the literature classifies as severe (Schanzenbach, 2015). We obtain these results by estimating the model separately for each crime category. Panel B of Table E1 reports the estimates.

We find particularly large minority penalties for drug crimes. In China, drug crimes are

[4] Among Han defendants in the respective baseline estimation samples, mean sentence length is 16.89 months, and the rates of life imprisonment and the death penalty are 1.300% and 0.405%, respectively. The latter two samples include only offenses eligible for the corresponding penalty.

[5] We assume records with missing ethnicity have the observed minority share and excluded defendants face the estimated disparities. Totals are provisional because counts and coefficients use different data versions.

a significant concern (Biddulph, 2002; Zhang and Chin, 2015), and ethnic minorities and minority criminal syndicates are disproportionately involved. For example, minority-dominated villages in Yunnan Province, which borders Myanmar—a country that supplies more than 90% of the heroin in Chinese markets—serve as important centers of drug trafficking. For drug crimes, ethnic minority defendants receive 5.3% longer sentences and are 11.2% and 19.2% more likely to receive life imprisonment and the death penalty, respectively (Figure 2, 3. Drug Crime).

4.2.3 Fact 3: Harsher penalties for less assimilated ethnic groups

Ethnic minority groups that are less assimilated into Han culture receive significantly harsher penalties in China's criminal justice system. Figure 2 reveals sizable additional penalties for ethnic groups that practice distinct religions. Specifically, the Chinese government views Christianity and Islam as potential political rivals because of their foreign origins, distinct cultural and linguistic traditions, and their own "spiritual leaders." Defendants from Christian ethnic groups (7) are 8.6% and 18.5% more likely than observably similar Han defendants to receive life imprisonment and the death penalty, respectively. Defendants from Muslim ethnic groups (8) are 6.2% and 16.6% more likely to receive these sentences. By contrast, defendants from ethnic groups that practice other religions, primarily Taoism, or no religion (9) face smaller increases of 2.5% and 5.8% in the likelihoods of receiving life imprisonment and the death penalty, respectively. Panel C of Table E1 reports the estimates.

We also find that ethnic minority groups with a history of conflict with the Han majority receive significantly more severe penalties. Their likelihoods of receiving life imprisonment and the death penalty increase by as much as 9.2% and 25.0%, respectively (10 in Figure 2).

Figure E1 further reports ethnicity-specific minority penalties among the 20 largest ethnic groups, with ethnic groups ordered by the magnitude of the estimated disparity. Although the estimates are somewhat noisy, presumably because some groups have relatively few cases, we find that the Dai, Tibetan, and Yi groups face the largest penalties in sentence length (Panel A); the Dai, Hani, and Yi groups face the largest penalties in life sentences (Panel B); and the Dai, Yi, and Yao groups face the largest penalties in death sentences (Panel C). The Dai ethnic group is concentrated in Yunnan Province, a major drug-trafficking hub in China, which may partly explain the large sentencing disparities observed among Dai defendants. These patterns are consistent with the broader results for groups that practice Christianity, such as the Yi and Hani, or Islam, such as the Hui, and for groups with a history of conflict with the Han majority, such as the Tibetans and Yi.

We find possible favoritism toward Uyghur defendants (Figure E2, Panels A–C), but this finding should be interpreted with caution. Uyghurs account for 0.81% of China's population in 2015 but only 0.08% of defendant observations in our data (Panel D). This gap raises

concerns about selective verdict publication, as politically sensitive cases may be excluded from CJO. Alternatively, it may reflect differences in how frequently Uyghur individuals appear in court and receive criminal sentences. Our data cannot distinguish between these explanations.

4.2.4 Fact 4: Larger minority penalties in areas with greater social tensions

Figure 2 shows that ethnic minority defendants face larger additional penalties in prefectures with lower incomes (13), larger ethnic minority populations (14), and higher crime rates (16).[6] These regional characteristics are closely correlated. Areas with larger ethnic minority populations are concentrated in western and southern China, where incomes are lower and crime rates are higher. These areas may also face a greater risk of social unrest because larger ethnic minority populations reduce the cost of collective action and lower incomes reduce the opportunity cost of participation (Cantoni et al., 2019). Panel D of Table E1 reports the estimates.

4.2.5 Fact 5: Han judges account for most disparities

Han judges predominantly drive the minority penalty (Figure E3). Across all three outcomes, Han judges impose substantially larger additional penalties on ethnic minority defendants than do ethnic minority judges. The estimated disparities are approximately three times as large among Han judges. For example, the estimated disparity in sentence length is 7.1% among Han judges and 2.8% among ethnic minority judges (column (2) of Table E1). Because Han judges account for 91% of judges, close to their share of the national population, they account for most of the observed minority penalty. Nevertheless, ethnic minority judges also impose harsher penalties on ethnic minority defendants, although the additional penalties are smaller. Panel E of Table E1 reports the estimates.

4.3 The Role of Prosecutors in Ethnic Sentencing Disparities

We supplement our analysis by examining prosecutors, who are also key actors in the criminal justice system, because judges alone may not account for the entire minority penalty in case outcomes (Rehavi and Starr, 2014; Tuttle, 2021; Baron et al., 2024). Prosecutors can recommend sentences to the court, and these recommendations may reflect bias in evidence collection or charging (Bohren et al., 2025). Our data report these recommendations for

[6]For each characteristic, we divide prefectures into high and low groups at the median and estimate the baseline equation separately for each group. Income and ethnic minority shares are measured using the 2010 census, and crime rates are calculated using the number of cases in our dataset.

about 16% of cases (see Appendix B-1 for details). We use them as the dependent variables in equation (1).

We find that prosecutors recommend harsher sentences for ethnic minority defendants than for observably similar Han defendants, although these disparities are smaller than those in final sentences. Figure E3 shows that ethnic minority defendants receive recommended prison terms that are 3.0% longer and are 3.5% and 8.5% more likely to receive recommendations for life imprisonment and the death penalty, respectively. These findings indicate that ethnic minority defendants are treated less favorably at multiple stages of the criminal process. Panel F of Table E1 reports the estimates.

5 Uyghur Unrest, Minority Population, and Minority Penalties

5.1 A Difference-in-Differences-in-Differences Approach

This section tests whether courts impose a larger minority sentencing penalty when the perceived threat of ethnic unrest increases. Following Wen (2024), we use conflict events involving Uyghur separatists and the Chinese government in Xinjiang as shocks to the nationwide salience of ethnic unrest. We expect the perceived threat of unrest to increase during conflict periods, particularly in locations with larger minority populations.

We estimate a difference-in-differences-in-differences (DDD) specification that combines variation across defendants, locations, and time. Specifically, we compare minority and Han defendants, prefectures above and below the median minority population share, and year-months with and without a conflict event in Xinjiang:

$$\begin{aligned} Sentence_{icpjt} = \alpha &+ \beta_1 Minor_i \times Unrest_t \times MinorPop_p \\ &+ \beta_2 Minor_i + \beta_3 Minor_i \times Unrest_t \\ &+ \beta_4 Minor_i \times MinorPop_p + \beta_5 Unrest_t \times MinorPop_p \\ &+ X_{icpjt} + \lambda_c + \sigma_j + \delta_t + \epsilon_{icpjt}. \end{aligned} \tag{2}$$

Here, $Unrest_t$ is an indicator equal to one if at least one conflict event occurs in Xinjiang in year-month t. $MinorPop_p$ is an indicator equal to one if the minority population share in prefecture p exceeds the sample median. We use the total minority population share, rather than the Uyghur population share, because Uyghurs constitute less than 1% of the population outside Xinjiang, unlike in the setting examined by Wen (2024). The two-way interaction terms allow both the minority–Han sentencing gap and the association between conflict periods and sentencing outcomes to differ between high- and low-minority-share prefectures. The year-month fixed effects absorb the main effect of $Unrest_t$, while the court fixed effects absorb the main effect of $MinorPop_p$.

Our coefficient of interest, β_1, measures whether the minority–Han sentencing gap in-

creases more during conflict periods in high-minority-share prefectures than in low-minority-share prefectures. A positive β_1 would therefore indicate that minority defendants receive relatively harsher sentences when concerns about ethnic unrest are heightened and that this response is stronger in prefectures where ethnic issues are likely to be more salient. Identification requires that, in the absence of the Xinjiang conflict events, the difference in the minority–Han sentencing gap between these two groups of prefectures would have evolved similarly during the corresponding months. However, a conventional event-study analysis is not feasible because the conflict events are few and occur consecutively within a short period. We cluster standard errors at the prefecture level.

5.2 Judicial Responses to Ongoing Unrest

We find that courts impose harsher penalties on ethnic minorities in response to heightened threats of unrest. The coefficient on the triple interaction ($Minor_i \times Unrest_t \times MinorPop_p$) is 0.301. This estimate indicates that, during periods of Uyghur unrest, the minority–Han gap in sentence length increases by an additional 0.301 months in high-minority-share prefectures relative to low-minority-share prefectures. This increase is about 0.24 times the baseline minority penalty estimated over the same period.[7] The coefficients for life imprisonment and the death penalty are 0.031 (Column (5)) and 0.025 (Column (6)), respectively, corresponding to approximately 37% and 44% of the respective baseline disparities in the full sample.

The effect on sentence length is driven primarily by working-age men. Following Wen (2024), we estimate the specification separately by defendants' gender and age in columns (2)–(4) of Table 1. The DDD coefficient for men aged 20–59 is 0.344 (Column (2)), similar to the full-sample estimate. In contrast, the coefficients for men aged 60 or older (Column (3)) and women (Column (4)) are statistically indistinguishable from zero, with point estimates of 0.001 and −0.020, respectively.[8] This pattern is consistent with courts responding more strongly to defendants perceived as more likely to participate in violence or collective action. Minority men account for approximately 4% of China's population but more than 45% of political prisoners in the CECC database (NBSC, 2000; CECC, 2019). Our findings, together with those of Wen (2024), suggest that China's local governments use both carrots and sticks to manage unrest. When the risk of conflict spreading rises, state firms hire more minority men (Wen, 2024), while local courts impose harsher sentences on this group.

[7] We re-estimate equation (1) using the same 3,131,373 defendant observations as the unrest regression, covering 2013–2017 and excluding Xinjiang. The baseline minority coefficient is 1.256 months (standard error: 0.066, clustered by court). The unrest coefficient is 0.301 months (standard error: 0.048, clustered by prefecture). The ratio is $0.301/1.256 \approx 0.24$.

[8] We do not conduct the corresponding subgroup analyses for life imprisonment or the death penalty because fewer than 600 older male defendants in our sample received these sentences. We also cannot examine the extensive margin of conviction because courts convict more than 99.9% of defendants.

6 Local Leaders' Career Incentives and Judicial Sentencing

6.1 A Difference-in-Differences-in-Differences Approach

Prefectural Party secretaries and mayors are evaluated on economic development, social stability, and other dimensions of local performance. Serious failures in stability maintenance can preclude promotion under the "one-vote veto" system, even when performance on other objectives is strong. Leaders with stronger promotion incentives may therefore place greater emphasis on preventing unrest and encourage courts to impose harsher penalties on ethnic minorities (de Janvry et al., 2023).

We test this hypothesis using China's age-dependent promotion system. Prefectural leaders are generally expected to "step back from frontline duties" at age 55, before formal retirement. Those who take office before this threshold generally retain opportunities for promotion, whereas those who take office afterward face sharply diminished prospects for advancement (Yao and Zhang, 2015; Yu et al., 2016; Campante et al., 2023).

We estimate DDD specifications analogous to those in Section 5.1:

$$\begin{aligned} Sentence_{icpjt} = \alpha &+ \beta_1 Minor_i \times Motivated_{pt} \times MinorPop_p \\ &+ \beta_2 Minor_i + \beta_3 Motivated_{pt} \\ &+ \beta_4 Minor_i \times Motivated_{pt} + \beta_5 Minor_i \times MinorPop_p \\ &+ \beta_6 Motivated_{pt} \times MinorPop_p + X_{icpjt} + \lambda_c + \sigma_j + \delta_t + \epsilon_{icpjt}. \end{aligned} \tag{3}$$

The variable $Motivated_{pt}$ equals one if at least one of the two principal leaders in prefecture p—the mayor or the party secretary—was younger than 55 at the beginning of their tenure, an age associated with a substantially greater likelihood of promotion. All other variables are defined as in equation (2). The coefficient β_1 measures whether the minority–Han sentencing gap widens more under promotion-motivated leaders in prefectures with larger minority shares. Note that the limited number of switches in leaders' promotion-incentive status prevents us from constructing adequate not-yet-treated comparison groups for an event-study analysis.

6.2 The Impact of Local Governments' Political Motivations on the Judiciary

We find that penalties against minority defendants become harsher when local prefectural leaders have greater opportunities for promotion and may therefore have stronger incentives to maintain local social order. In Table 2, our interest of coefficient is 0.09 for sentence length (Column (1)), corresponding to 8% of our baseline results. This means that switching to politically motivated leaders in minority-dominant prefectures increases the sentence length for ethnic minorities by 8%, relative to the corresponding impact in non-minority prefectures. The coefficients for life imprisonment and the death penalty are 0.017 and 0.016,

respectively, corresponding to increases of 20% and 28% relative to the respective baseline minority penalties, although both estimates are imprecise. These estimates capture sizable additional penalties associated with stronger promotion incentives, even though leaders with limited promotion prospects remain responsible for maintaining social stability.

7 Conclusion

Using newly digitized data covering nearly the universe of published criminal verdicts in China, we provide the first comprehensive nationwide analysis of ethnic sentencing disparities. We find that ethnic minorities receive disproportionately harsher penalties, with 6.8% longer sentences and 6.5% and 14.0% higher probabilities of life imprisonment and the death penalty, respectively, compared to observably similar majority defendants. These additional penalties are particularly severe for crimes associated with social unrest, among less assimilated ethnic groups, and in regions experiencing heightened social tensions. The sentence-length penalty against minorities increases in prefectures with larger minority populations during Uyghur unrest and under local leaders with stronger promotion incentives, suggesting political roots of the disparity.

These estimates require caution for three reasons. First, our estimates alone cannot establish that discrimination drives the sentencing gap, as unobserved factors correlated with ethnicity may also explain it. Nonetheless, if courts impose harsher penalties because they associate a defendant's ethnicity with a greater risk of unrest, such decisions constitute discrimination and conflict with the principle of equal treatment under the rule of law. Second, we do not account for selection into arrest and prosecution; such data are, to our knowledge, extremely challenging to obtain in China. Third, our data omit some politically sensitive cases and therefore provide an incomplete picture of how courts treat ethnic minorities.

Our results suggest political roots of ethnic sentencing disparities. Previous studies offer two common explanations for sentencing disparities, but neither fully explains our findings. First, our results do not support in-group favoritism as the sole explanation (Shayo and Zussman, 2011; Anwar et al., 2012; Ash et al., 2021). In-group favoritism does not explain why additional penalties concentrate in crimes and ethnic groups associated with perceived threats to social order. Minority judges also impose harsher penalties on minority defendants, seemingly contradicting in-group favoritism. Second, U.S. studies suggest that local courts respond to voters' preferences (Lim, 2013; Berdejo and Yuchtman, 2013; Feigenberg and Miller, 2021). In China, however, minority penalties are larger where minorities make up a greater share of the local population, and they widen when local leaders face stronger promotion incentives. These patterns suggest a distinct political role for the Chinese judiciary: supporting the Party's goal of preventing social unrest.

Our interpretation raises a puzzle. Since 2017, international observers have raised concerns about increasingly restrictive ethnic policies in Xinjiang (OHCHR, 2022). If courts help maintain state control, we might therefore expect ethnic sentencing disparities to widen. Our results show the opposite. Figure 3 shows that the sentence-length gap generally narrowed. By 2021, the gaps in sentence length and in the probabilities of life imprisonment and the death penalty were statistically indistinguishable from zero.

One potential explanation for the near disappearance of the ethnic sentence-length gap is China's national effort to improve the transparency and consistency of judicial decisions. These efforts include sentencing guidelines (2010, 2014, 2017), rules governing the use of guiding cases (2015), safeguards against officials' interference in individual cases (2015), online judgment-publication requirements (2016), and requirements to explain judicial reasoning (2018) (SPC, 2015b,a, 2016, 2018). Among these measures, sentencing guidelines directly address how judges determine sentence lengths. China introduced these guidelines on a nationwide trial basis in 2010, formally implemented them in 2014, and revised them and piloted their expansion to additional offenses in 2017 (Guo, 2014; Chen, 2017). The guidelines limit judges' discretion (SPC, 2014). For example, the criminal law prescribes at least seven years for trafficking either 20 or 25 grams of heroin, without specifying how much the additional five grams should increase the sentence. The guidelines specify a one-year increase in the baseline sentence (Jiangsu HPC, 2014). By limiting judicial discretion, these rules may reduce courts' scope to impose harsher penalties on ethnic minorities.

Evaluating the effects of these reforms is challenging. The Supreme People's Court began requiring courts to publish judgments online in 2014. Therefore, we cannot credibly estimate the effects of the 2010 and 2014 reforms without sufficient pre-reform data. We also find no statistically significant effect on the ethnic sentencing gap from the 2017 reform, which revised existing guidelines and piloted their extension to additional offenses, rather than introducing guidelines for the first time.

Regardless of its causes, the remarkable decline in ethnic sentencing disparities suggests progress toward more equal treatment in China's courts. Further research is needed to understand what drives this decline and how to promote fairer sentencing.

References

Adena, M., Enikolopov, R., Petrova, M., Santarosa, V., and Zhuravskaya, E. (2015). Radio and the rise of the nazis in prewar germany. *The Quarterly Journal of Economics*, 130(4):1885–1939.

Al Jazeera (2018). One million muslim uighurs held in secret china camps: UN panel. *Al Jazeera*.

Andirin, V., Neggers, Y., Shadmehr, M., and Shapiro, J. M. (2022). Surveillance of repression: Theory and implementation. Working Paper 30167, National Bureau of Economic Research. Revised March 2025.

Anwar, S., Bayer, P., and Hjalmarsson, R. (2012). The impact of jury race in criminal trials. *The Quarterly Journal of Economics*, 127(2):1017–1055.

Arnold, D., Dobbie, W., and Hull, P. (2022). Measuring racial discrimination in bail decisions. *American Economic Review*, 112(9):2992–3038.

Ash, E., Asher, S., Bhowmick, A., Bhupatiraju, S., Chen, D., Devi, T., Goessmann, C., Novosad, P., and Siddiqi, B. (2021). In-group bias in the indian judiciary: Evidence from 5 million criminal cases. *URL: https://www. devdatalab. org/judicial-data*.

Ash, E., Chen, D. L., and Naidu, S. (2026). Ideas have consequences: The impact of law and economics on American justice. *The Quarterly Journal of Economics*, 141(1):845–887.

Baron, E. J., Doyle, Jr., J. J., Emanuel, N., Hull, P., and Ryan, J. (2024). Discrimination in multiphase systems: Evidence from child protection. *The Quarterly Journal of Economics*, 139(3):1611–1664.

BBC News (n.d.). Xinjiang. BBC News.

Beraja, M., Kao, A., Yang, D. Y., and Yuchtman, N. (2023). Ai-tocracy. *The Quarterly Journal of Economics*, 138(3):1349–1402.

Berdejo, C. and Yuchtman, N. (2013). Crime, punishment, and politics: an analysis of political cycles in criminal sentencing. *Review of Economics and Statistics*, 95(3):741–756.

Bhuller, M., Dahl, G. B., Løken, K. V., and Mogstad, M. (2020). Incarceration, recidivism, and employment. *Journal of Political Economy*, 128(4):1269–1324.

Biddulph, S. (2002). The production of legal norms: a case study of administrative detention in china. *UCLA Pac. Basin LJ*, 20:217.

Bohren, J. A., Hull, P., and Imas, A. (2025). Systemic discrimination: Theory and measurement. *The Quarterly Journal of Economics*, 140(3):1743–1799.

Campante, F. R., Chor, D., and Li, B. (2023). The political economy consequences of china's export slowdown. *Journal of the European Economic Association*, 21(5):1721–1771.

Cantoni, D., Yang, D. Y., Yuchtman, N., and Zhang, Y. J. (2019). Protests as strategic games: experimental evidence from hong kong's antiauthoritarian movement. *The Quarterly Journal of Economics*, 134(2):1021–1077.

CECC (2019). Political prisoner database. Congressional-Executive Commission on China.

Chen, X. (2017). Supreme people's court calls for expanding the scope of sentencing standardization. People's Court Daily, April 1. https://www.court.gov.cn/zixun/xiangqing/39112.html. In Chinese. Reproduced on the Supreme People's Court website.

Chen, Y. and Yang, D. Y. (2019). The impact of media censorship: 1984 or brave new world? *American Economic Review*, 109(6):2294–2332.

CJO (2021). China judgments online. Accessed: [2017-2021].

Cohen, A. and Yang, C. S. (2019). Judicial politics and sentencing decisions. *American Economic Journal: Economic Policy*, 11(1):160–191.

Costa, D. L. and Zhao, Z. (2025). Equality before the law in U.S. Civil War Courts-Martial. Working Paper 34184, National Bureau of Economic Research.

de Janvry, A., He, G., Sadoulet, E., Wang, S., and Zhang, Q. (2023). Subjective performance evaluation, influence activities, and bureaucratic work behavior: Evidence from China. *American Economic Review*, 113(3):766–799.

Egorov, G. and Sonin, K. (2024). The political economics of non-democracy. *Journal of Economic Literature*, 62(2):594–636.

Feigenberg, B. and Miller, C. (2021). Racial divisions and criminal justice: Evidence from southern state courts. *American Economic Journal: Economic Policy*, 13(2):207–240.

Gehlbach, S., Sonin, K., and Svolik, M. W. (2016). Formal models of nondemocratic politics. *Annual Review of Political Science*, 19(1):565–584.

Guo, S. (2014). Sentencing standardization: Toward greater standardization and transparency. People's Court Daily, March 17. https://www.court.gov.cn/zixun/xiangqing/6227.html. In Chinese. Reproduced on the Supreme People's Court website.

Guriev, S. and Treisman, D. (2019). Informational autocrats. *Journal of economic perspectives*, 33(4):100–127.

Hochreiter, S. and Schmidhuber, J. (1997). Long short-term memory. *Neural computation*, 9(8):1735–1780.

Hou, Y. and Truex, R. (2022). Ethnic discrimination in criminal sentencing in china. *The Journal of Politics*, 84(4):2294–2299.

Human Rights Watch (n.d.). Uyghurs. Human Rights Watch.

Jiangsu High People's Court (2014). Implementing rules for the guiding opinions on sentencing for common crimes. https://www.jsfy.gov.cn/article/68607.html. In Chinese. Official text published August 20.

King, G., Pan, J., and Roberts, M. E. (2013). How censorship in china allows government criticism but silences collective expression. *American political science Review*, 107(2):326–343.

Lai, H. (2009). China's ethnic policies and challenges. *East Asian Policy*, 1(3):5–13.

Li, L. (2016). The chinese communist party and people's courts: Judicial dependence in china. *The American Journal of Comparative Law*, 64(1):37–74.

Li, Y., Longmire, D., and Lu, H. (2018). Death penalty disposition in China: What matters? *International Journal of Offender Therapy and Comparative Criminology*, 62(1):253–273.

Lim, C. S. H. (2013). Preferences and incentives of appointed and elected public officials: Evidence from state trial court judges. *American Economic Review*, 103(4):1360–1397.

Lin, T.-Y., Goyal, P., Girshick, R., He, K., and Dollár, P. (2017). Focal loss for dense object detection. In *Proceedings of the IEEE international conference on computer vision*, pages 2980–2988.

Liu, Z., Peng, W., and Wang, S. (2026). A few bad apples? academic dishonesty, political selection, and institutional performance in China. Working Paper 34747, National Bureau of Economic Research.

Martinez-Bravo, M., Padró i Miquel, G., Qian, N., and Yao, Y. (2022). The rise and fall of local elections in china. *American Economic Review*, 112(9):2921–2958.

National Bureau of Statistics of China (2000). 2000 population census of china. Census data.

Ng, K. H. and He, X. (2017). *Embedded Courts: Judicial Decision-Making in China*. Cambridge University Press, Cambridge.

Office of the United Nations High Commissioner for Human Rights (2022). OHCHR assessment of human rights concerns in the xinjiang uyghur autonomous region, people's republic of china. United Nations.

Olson, M. (1993). Dictatorship, democracy, and development. *American political science review*, 87(3):567–576.

Peng, Y. and Cheng, J. (2022). Ethnic disparity in chinese theft sentencing. *China Review*, 22(3):47–71.

Pereira, A. W. (2005). *Political (in) justice: authoritarianism and the rule of law in Brazil, Chile, and Argentina*. University of Pittsburgh Press.

Rehavi, M. M. and Starr, S. B. (2014). Racial disparity in federal criminal sentences. *Journal of Political Economy*, 122(6):1320–1354.

Rozenas, A. and Stukal, D. (2019). How autocrats manipulate economic news: Evidence from russia's state-controlled television. *The Journal of Politics*, 81(3):982–996.

Sautman, B. (1998). Preferential policies for ethnic minorities in china: The case of xinjiang. *Nationalism and Ethnic Politics*, 4(1–2):86–118.

Schanzenbach, M. M. (2015). Racial disparities, judge characteristics, and standards of review in sentencing. *Journal of Institutional and Theoretical Economics: JITE*, pages 27–47.

Shayo, M. and Zussman, A. (2011). Judicial ingroup bias in the shadow of terrorism. *The Quarterly journal of economics*, 126(3):1447–1484.

Solomon, P. H. (1996). *Soviet criminal justice under Stalin*, volume 100. Cambridge University Press.

Steinert, C. V. (2023). The duration of political imprisonment: Evidence from China. *Conflict Management and Peace Science*, 40(4):349–372.

Supreme People's Court (2014). Guiding opinions on sentencing for common crimes. https://www.court.gov.cn/shenpan/xiangqing/6622.html. In Chinese. Official text published July 31.

Supreme People's Court (2015a). Implementing measures for recording, reporting, and accountability for officials' interference in judicial activities and individual cases. https://www.court.gov.cn/fabu/xiangqing/15206.html. In Chinese. Effective August 20.

Supreme People's Court (2015b). Implementing rules for the provisions on case guidance. https://www.hncourt.gov.cn/public/detail.php?id=157832. In Chinese. Issued May 13; reproduced by the Henan High People's Court.

Supreme People's Court (2016). Provisions on the publication of judgments by people's courts on the internet. https://www.court.gov.cn/zixun/xiangqing/25321.html. In Chinese. Effective October 1.

Supreme People's Court (2018). Guiding opinions on strengthening and standardizing reasoning in judicial documents. https://www.court.gov.cn/zixun/xiangqing/101552.html. In Chinese. Effective June 13.

Supreme People's Court et al. (2020). Opinions on several issues concerning the regulation of sentencing procedures. Supreme People's Procuratorate of China.

Tian, J. (2023). The fairness of sentencing for ethnic minorities: An empirical study on drug crime and sentencing in China. *Journal of Sociology and Ethnology*, 5(3):67–79.

Tuttle, C. (2021). Racial disparities in federal sentencing: Evidence from drug mandatory minimums.

Wen, J. (2024). State employment as a strategy of autocratic control in china. *Working Paper*.

Wikipedia contributors (n.d.). Xinjiang conflict. Wikipedia.

Woo, M. Y. K. (1999). Law and discretion in the contemporary chinese courts. *Pacific Rim Law & Policy Journal*, 8(3):581–616.

Xu, X. (2021). To repress or to co-opt? authoritarian control in the age of digital surveillance. *American Journal of Political Science*, 65(2):309–325.

Yanagizawa-Drott, D. (2014). Propaganda and conflict: Evidence from the rwandan genocide. *The Quarterly Journal of Economics*, 129(4):1947–1994.

Yao, Y. and Zhang, M. (2015). Subnational leaders and economic growth: Evidence from chinese cities. *Journal of Economic Growth*, 20:405–436.

Yu, J., Zhou, L.-A., and Zhu, G. (2016). Strategic interaction in political competition: Evidence from spatial effects across chinese cities. *Regional Science and Urban Economics*,

57:23–37.

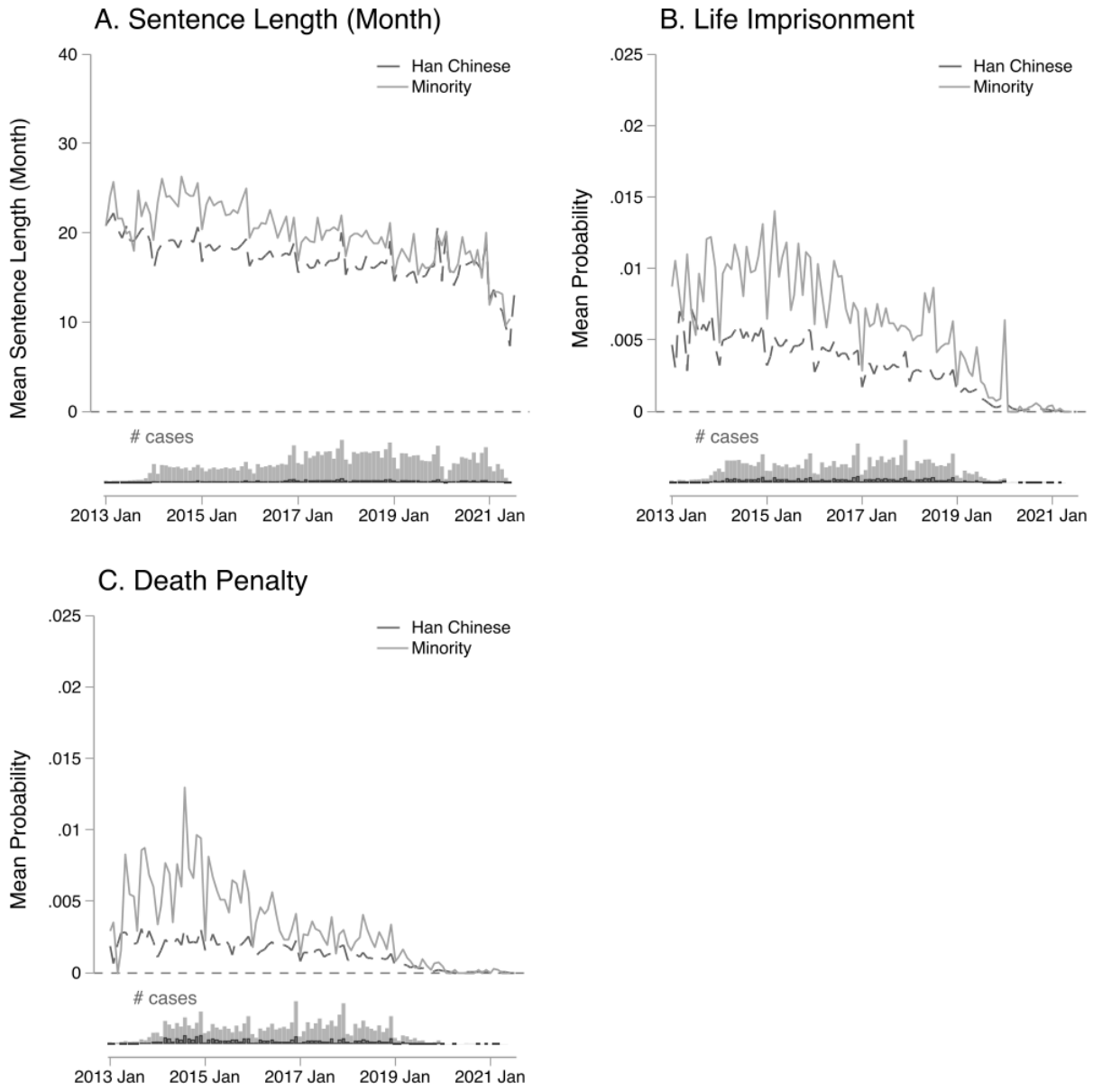
Zhang, H. (2011). The laws on the ethnic minority autonomous regions in china: Legal norms and practices. *Loy. U. Chi. Int'l L. Rev.*, 9:249.

Zhang, S. X. and Chin, K.-l. (2015). A people's war: China's struggle to contain its illicit drug problem. *Brookings Institution*, 1.

Zhao, H. (2016). Party-member judges must strengthen their commitment to justice for the people. Legal Daily.

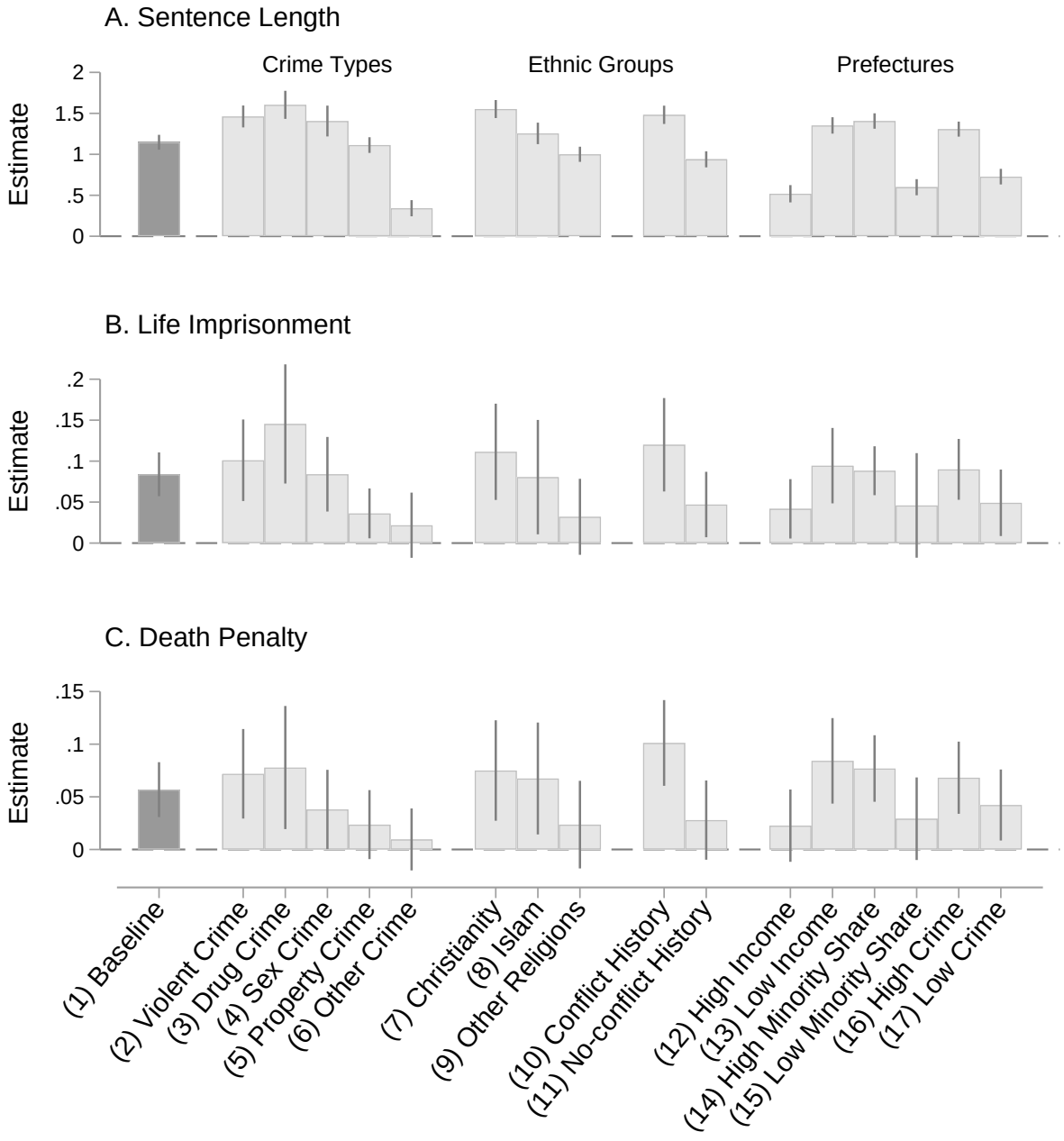
Figures

Figure 1: Time Trend of Average Sentencing Between Han and Minority



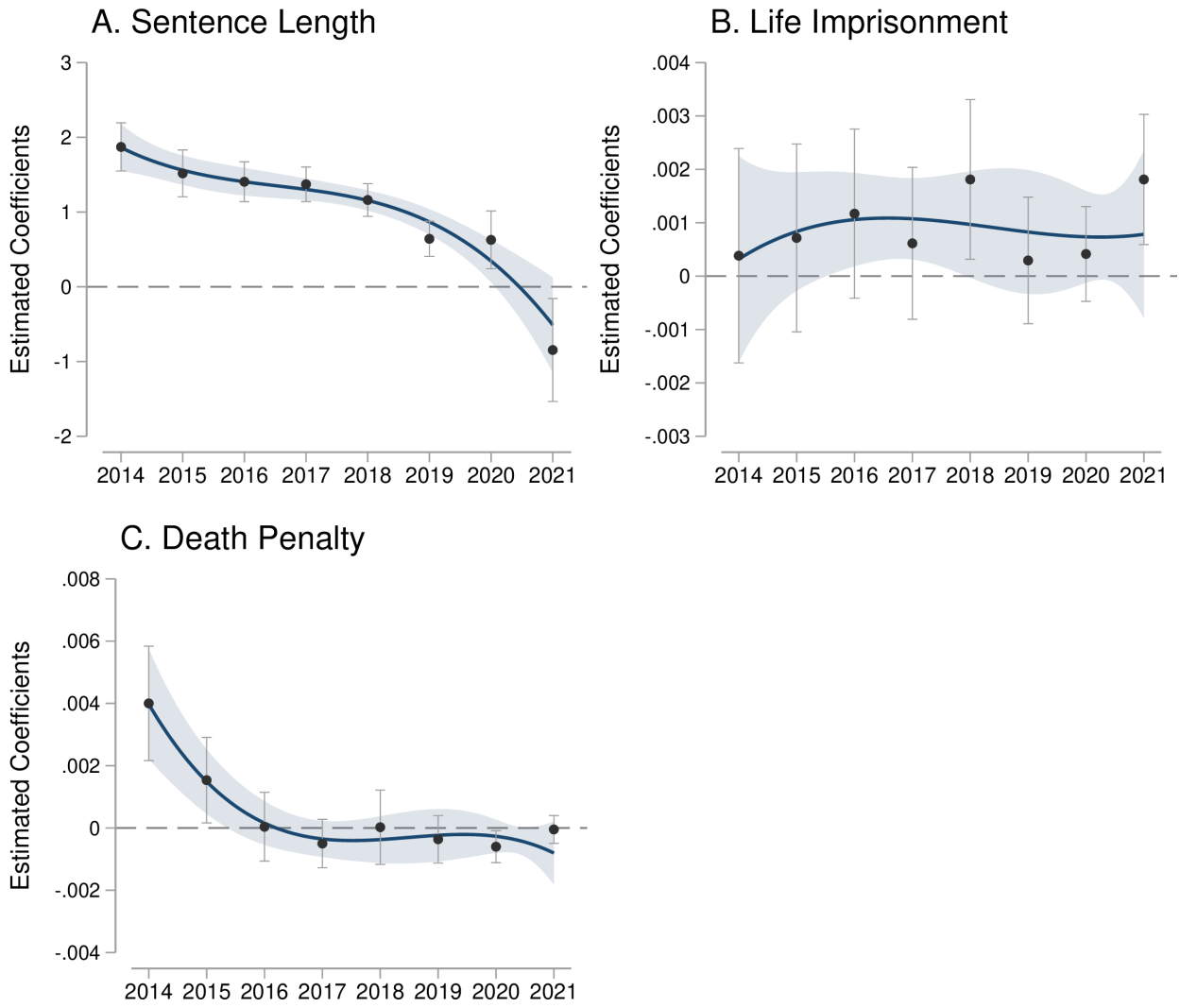
Notes: This figure illustrates the time trend of average sentencing outcomes for Han and Minority groups. Panel (a) displays the average imprisonment sentence length. Panel (b) shows the average probability of life imprisonment. Panel (c) presents the average probability of receiving a death sentence. The bar below represents the number of corresponding cases, with darker gray indicating Minority cases and lighter gray indicating Han majority cases.

Figure 2: Facts about Ethnic Minority Issues in China's Criminal Justice System



Notes: This figure presents key findings on the facts about ethnic minority issues in China's criminal justice system. The outcome variables are sentence length in months (Panel A), life imprisonment (Panel B), and the death penalty (Panel C). For life imprisonment and the death penalty, we restrict the samples to crime types that are eligible for these sentencing penalties. The bars denote the estimated coefficients, while the lines represent the 95% confidence intervals. The category labels below Panel C apply to all three panels. We cluster standard error at the court level. See the main text for details. Table E1 reports the estimates and standard errors. Sentence-length coefficients are in months; life-imprisonment and death-penalty coefficients are in percentage points.

Figure 3: Minority Penalty over Time



Notes: This figure reports the estimated minority–Han sentencing gap over 2013–2021. Panel A reports sentence length in months, Panel B reports the probability of life imprisonment, and Panel C reports the probability of the death penalty. The points show coefficients from regressions that interact minority status with year indicators; whiskers show their 95% confidence intervals. The colored lines show cubic time trends estimated from individual-level regressions, and the shaded areas show their pointwise 95% confidence intervals. The years 2013 and 2014 are pooled into one initial period in both specifications. All specifications include the replication controls and fixed effects, including actual year-month fixed effects. The life-imprisonment and death-penalty samples are restricted to eligible offenses. The 2021 estimates use the available partial-year observations. Standard errors are clustered at the court level.

Tables

Table 1: Uyghur Unrest and Penalties against Ethnic Minorities

	Sentence length (months)					
	All	Men age 20–59	Men age 60+	Women	Life imprison-ment	Death penalty
	(1)	(2)	(3)	(4)	(5)	(6)
Minority × Unrest × Minority Pref	0.301***	0.344***	0.001	-0.020	0.031**	0.025*
	(0.048)	(0.076)	(0.074)	(0.063)	(0.015)	(0.014)
Observations	3,131,373	1,976,305	55,818	261,122	877,448	1,087,269
Crime category FE	Y	Y	Y	Y	Y	Y
Case characteristics	Y	Y	Y	Y	Y	Y
Defendant characteristics	Y	Y	Y	Y	Y	Y
Year-month FE	Y	Y	Y	Y	Y	Y
Province FE	Y	Y	Y	Y	Y	Y
Court FE	Y	Y	Y	Y	Y	Y
Judge FE	Y	Y	Y	Y	Y	Y

Notes: This table reports the DDD estimates from equation (2). Columns (1)–(4) report sentence-length estimates in months for all defendants, men aged 20–59, men aged 60 or older, and women, respectively. Columns (5) and (6) include all demographic groups and restrict the sample to crimes eligible for life imprisonment and the death penalty, respectively. Minority identifies ethnic-minority defendants; Unrest identifies months with a Uyghur conflict event; Minority Pref identifies prefectures at or above the configured minority-population-share cutoff. All specifications use cases from 2013–2017 and exclude Xinjiang. The relevant lower-order interactions enter every specification. Standard errors, in parentheses, are clustered by prefecture. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table 2: Local Governments, Judiciary, and Penalties against Minorities

	Sentence Length	Life Imprison -ment	Death Penalty
	(1)	(2)	(3)
Minority × Below Age 55 × Minority Pref	0.091***	0.017	0.016
	(0.032)	(0.075)	(0.069)
Crime category FE	Y	Y	Y
Case characteristics	Y	Y	Y
Defendant characteristics	Y	Y	Y
Year-month FE	Y	Y	Y
Province FE	Y	Y	Y
Court FE	Y	Y	Y
Judge FE	Y	Y	Y

Notes: This table presents the DDD results from equation (3). "Minority" indicates whether the defendant belongs to an ethnic minority group. "Below Age 55" equals one if at least one of the two prefectural leaders—the mayor or Party secretary—was younger than 55 at the start of their tenure. "Minority Pref" equals one if the prefectural minority population share is above the median. See the main text for further details. Standard errors, clustered at the prefecture level, are reported below the coefficients. * significant at 10%; ** significant at 5%; *** significant at 1%.

ONLINE APPENDIX

Authoritarian Control through the Judiciary

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A Additional Background

A-1 Criminal Procedure

China's criminal justice system assigns distinct but coordinated roles to public security organs, People's Procuratorates, and People's Courts. Police investigate suspected offenses, gather evidence, and transfer sufficiently supported cases to the procuratorate. Prosecutors review the evidence and legality of the investigation, then initiate prosecution, decline to prosecute, or request supplementary investigation. Cases prosecuted proceed to court for adjudication. Our analysis examines ethnic disparities arising after individuals enter this system.

Criminal cases are adjudicated within a four-level court hierarchy, with most first-instance trials conducted by primary or intermediate people's courts. China follows the principle that the second instance is final: a first-instance judgment becomes final unless the defendant appeals or the procuratorate files a protest. When either occurs, a higher-level court reviews the case, and its judgment is generally final.

Two features of adjudication are particularly relevant to our analysis. First, 99.994% of defendants whose cases reached adjudication in 2018 were convicted. Within the court verdicts observed in our data, the principal variation therefore concerns sentence severity rather than conviction or acquittal. Second, judges determine the relevant facts and apply statutory sentencing rules while retaining considerable discretion within prescribed ranges and in applying aggravating or mitigating provisions. Drug offenses illustrate this discretion: statutory ranges depend on drug type and quantity, while the precise sentence also depends on other circumstances of the offense.

China's treatment of guilty pleas also matters for sentencing. Unlike the negotiated plea bargaining prevalent in the United States, China's system of leniency for confession and acceptance of punishment allows defendants to agree to prosecutorial sentencing recommendations while requiring court adjudication. Defendants who confess and accept punishment generally receive more lenient sentences, and use of this system has expanded in recent years.[1] We account for this variation through a corresponding control variable.

[1] See https://www.chinalawtranslate.com/en/a-plea-for-greater-reform-and-engagement/.

B Data Appendix

B-1 Variable Construction

We explain the detailed variables used in our analysis. First, we extract institutional and demographic characteristics. The judgments identify the court and judge responsible for each case, as well as the year and month of sentencing. Our data cover 3,506 courts and 49,187 judges. We also extract defendants' age, gender, ethnicity, education, occupation, and criminal history.

Second, we construct a detailed classification of criminal offenses. We classify cases into 427 distinct crime types. For example, we divide drug-related offenses into 13 categories and separately classify illegal possession, trafficking, production, cultivation, harboring drug offenders, and facilitating drug use.

Third, we extract case characteristics that criminal law recognizes as aggravating or mitigating sentencing factors. These include both crime-specific measures of severity and factors applicable across crimes. Crime-specific measures include the number of victims killed in homicide cases, the severity of injuries in assault cases, the types and quantities of drugs involved in drug-related offenses, and the value of stolen property in theft cases. Common factors include guilty pleas, voluntary surrender, forgiveness by the victim, and information on defendants' attitudes, motives, mental disorders, and disabilities. Table B1 provides examples of these controls.

Finally, we extract judicial outcomes and prosecutors' sentencing recommendations. Approximately 20% of crime types permit life imprisonment or the death penalty. We also observe prosecutors' recommended sentences in about 15.7% of cases, allowing us to examine whether ethnic disparities arise in prosecutorial recommendations or judicial sentencing decisions. Under the 2010 joint rules, recommendations are reported when the facts are clear, the evidence is credible and sufficient, and relevant sentencing factors have been considered (SPC et al., 2020).

B-2 Construction of Main Sample

We describe how we construct the sample. Since 2014, the Supreme People's Court of China has required courts at all levels to publish both newly issued and historical judgments through the platform, giving CJO broad coverage of criminal court decisions. The platform nevertheless removes large batches of judgments without public notice, including a major batch deletion in 2021. To limit the resulting data loss, we use an automated daily collection process that archives judgments before they are removed.

CJO does not provide complete coverage of Chinese court cases. The platform does not publish juvenile cases or certain politically sensitive cases. Although the Supreme People's

Court has required courts to publish judgments online since 2014, coverage varies across courts and years. Some courts also publish judgments issued before 2014, while others have incomplete coverage even after the requirement took effect. Our data contain fewer judgments from 2014–2016 than from 2017–2018. Despite these limitations, the raw CJO data contain 11.78 million judgments from 2014 to 2021, equivalent to approximately 89% of the cases reported by the Supreme People's Court.

We restrict our analysis to first-instance judgments, which account for 71% (8.4 million) of the raw data. The remaining records consist of sentence-commutation decisions (21%), second-instance judgments (6.1%), and modifications (1.9%). Excluding second-instance judgments entails little loss of independent information because more than 95% of these proceedings leave the original judgment unchanged.

The input dataset contains 8,769,103 records. We exclude 2,185,557 records with missing defendants' ethnicity and 142,585 additional records with missing sentence length, leaving 6,440,961 cases. Removing 6,816 singleton observations yields a final sentence-length estimation sample of 6,434,145 cases, or 73.4% of the input dataset. The life-imprisonment and death-penalty analyses use separate samples restricted to eligible offenses.

Table B3 reports summary statistics for the 6,583,546 records with known ethnicity, combining 5,946,526 Han and 637,020 ethnic minority defendants. Statistics use nonmissing observations for each variable. Table B2 compares the full input dataset with the sentence-length estimation sample. The samples are broadly similar in observed defendant and case characteristics, partly alleviating concerns about selection due to missing-data exclusions.

B-3 Descriptive Statistics and Sample Comparisons

Table B3 reports the summary statistics. Overall, we find that ethnic minority and Han defendants differ in various respects. Ethnic minority defendants tend to be younger and less educated and to earn less, while their criminal histories are similar (Panel A). During court proceedings, they are less likely to plead guilty, surrender voluntarily, receive forgiveness from their victims, or receive suspended sentences, while the severity of their offenses does not differ significantly between the two groups. Regarding sentencing outcomes, ethnic minority defendants receive, on average, substantially harsher penalties than Han defendants (Panel C). The unconditional mean length of confinement is 19.8 months for ethnic minority defendants and 16.9 months for Han defendants. This observed discrepancy is even greater for more severe offenses. The probabilities of receiving life and death sentences are 0.270% and 0.115%, respectively, for Han defendants, compared with 0.574% and 0.292% for ethnic minority defendants.

Table B1: Examples of Case Characteristics Used as Controls

Category	Examples
A. General Factors	
Culpability and personal circumstances	Negligence; mental illness or disability; repeat offending
Stage and role in the offense	Preparation; attempted or discontinued offenses; principal or secondary role; coercion
Cooperation and redress	Confession; voluntary surrender; useful information; remorse; compensation; victim forgiveness; settlement
Aggravating circumstances	Targeting vulnerable victims; offending during disasters; cruel methods; severe consequences
B. Crime-specific Factors	
Theft	Value stolen; burglary; pickpocketing; repeated theft; weapons
Drug trafficking	Drug type, quantity, and purity; armed trafficking; involvement of minors; international trafficking
Assault	Number of victims by injury severity; number of deaths
Homicide	Number of deaths

Notes: This table lists selected examples of the case characteristics used as controls. Sources: Criminal Law of the People's Republic of China (2011 Amendment); Guiding Opinions on the Sentencing Procedures of the People's Courts (2017 Version); Implementing a balanced criminal policy of severity and leniency in cases of intentional homicide, injury, and organized crime with mafia-like characteristics (《在审理故意杀人、伤害及黑社会性质组织犯罪案件中切实贯彻宽严相济刑事政策》, 2010).

Table B2: Balance Check: Comparing the Original Data and Main Sample

	All (N = 8,769,103)		Baseline sample (N = 6,434,145)		
	Mean	S.D.	Mean	S.D.	Dif
	(1)	(2)	(3)	(4)	(5)
A. Defendant characteristics:					
Age	37.6	(11.2)	37.6	(11.2)	0.033
Gender (male = 1)	.909	(.288)	.909	(.288)	-0.000
Primary school	.29	(.454)	.291	(.454)	-0.001
Secondary school	.481	(.5)	.483	(.5)	-0.003
High school or junior college	.198	(.398)	.196	(.397)	0.002
Above 4-year college	.031	(.173)	.0292	(.168)	0.002
B. Case characteristics					
Offense level 1	0.27	(.444)	0.276	(.447)	-0.005
Offense level 2	0.661	(.473)	.649	(.477)	0.012
Offense level 3	0.0687	(.253)	.0754	(.264)	-0.007
Guilty plea (yes = 1)	0.074	(.262)	.0843	(.278)	-0.010
Surrender (yes = 1)	.465	(.50)	.463	(.499)	0.002
Confession (yes = 1)	.696	(.460)	.70	(.458)	-0.004
Plaintiff Forgive (yes = 1)	.214	(.410)	.217	(.412)	-0.003
C. Case resolution					
Sentence length (month)	17.2	(25.7)	17.2	(25.7)	0.049
Life imprisonment (yes = 1)	.00312	(.0558)	.00000	(.0000)	0.003
Death penalty (yes = 1)	.0016	(.0404)	.0000	(.0000)	0.002

Notes: This table compares variables between all records supplied in Wenwei's data and those retained in the baseline sentence-length estimation sample. Column (5) shows the difference between Columns (1) and (3). See Appendix B-2 for details on how the baseline sample is constructed.

Table B3: Summary Statistics of Key Variables

	Han Chinese (N = 5,946,526)		Ethnic minority (N = 637,020)		All (N = 6,583,546)	
	Mean	S.D.	Mean	S.D.	Mean	S.D.
	(1)	(2)	(3)	(4)	(5)	(6)
A. Defendant characteristics:						
Age	37.9	(11.2)	35.0	(10.7)	37.7	(11.2)
Gender (male = 1)	.907	(.290)	.919	(.272)	.908	(.288)
Primary school and below	.276	(.447)	.422	(.494)	.290	(.454)
Secondary school	.488	(.500)	.414	(.493)	.481	(.500)
High school / junior college	.204	(.403)	.139	(.346)	.198	(.398)
Above 4-year college	.031	(.174)	.024	(.154)	.031	(.173)
High-paid Job (yes = 1)	.102	(.302)	.062	(.241)	.098	(.297)
B. Case characteristics						
Offense level (1-3)	1.789	(.568)	1.793	(.511)	1.790	(.563)
Crime History (yes = 1)	.167	(.373)	.172	(.378)	.167	(.373)
Guilty plea (yes = 1)	.086	(.280)	.054	(.225)	.083	(.275)
Surrender (yes = 1)	.464	(.499)	.429	(.495)	.461	(.498)
Confession (yes = 1)	.696	(.460)	.694	(.461)	.695	(.460)
Plaintiff Forgive (yes = 1)	.220	(.414)	.183	(.387)	.216	(.412)
C. Case resolution						
Sentence length (month)	16.9	(25.2)	19.8	(29.5)	17.2	(25.7)
Life imprisonment (yes = 1)	.003	(.052)	.006	(.076)	.003	(.055)
Death penalty (yes = 1)	.001	(.034)	.003	(.054)	.001	(.036)

Notes: This table presents summary statistics for records with observed Han/minority status. The All columns combine the Han and minority samples. Statistics use non-missing observations for each variable, so sample sizes vary across rows. Panel A reports the summary statistics for the personal characteristics of criminal defendants, Panel B covers the characteristics of criminal cases, and Panel C describes the sentencing outcomes.

C Verdict Example

The example below illustrates the information we extract from criminal judgments. The first section records defendant characteristics, the second describes the offense, and the third reports sentencing factors and penalties. We provide an English translation followed by the Chinese original.

C-1 English Translation

Part I

People's Court of Hepu County, Guangxi Zhuang Autonomous Region Criminal Judgment

Public prosecution agency: Hepu County People's Procuratorate, Guangxi Zhuang Autonomous Region.
Defendant Zhu Youchuan, a male who was born on July 12, 1984 in Hepu County, Guangxi, with Han ethnicity and junior high school education background. He was unemployed and lived in Hepu County, Guangxi Province when arrested. Due to drug abuse on January 17, 2018, he was administratively detained for 12 days by the Beihai City Public Security Bureau Yinhai Branch Office. On January 29, 2018, the Beihai City Public Security Bureau Yinhai Branch decided to conduct compulsory isolation and detoxification for two years for him. He was arrested on March 19, 2018, for being suspected of drug trafficking and was arrested on March 7, 2018, by the Beihai Municipal Public Security Bureau Yinhai Branch. He is currently detained at the Second Detention Center in Beihai City.

Part II

The Hepu County People's Procuratorate accused the defendant Zhu Youchuan of trafficking in dangerous drugs on the basis of the indictment of the Joint Public Prosecution [2018] No. 236, and filed a public prosecution on May 24, 2018. The court applied the simple procedure in accordance with the law, implemented a sole trial, and heard the case in public. The Hepu County People's Procuratorate appointed the procurator Pang Bin to appear in court to support the public prosecution. The defendant Zhu Youchuan went to court to participate in the lawsuit. It has now been accomplished.

After investigation, it was found that at around 14:00 on January 16th, 2018, Chen, a drug purchaser, contacted someone whose nickname is "Er Ge" to purchase 200 yuan of heroin and agreed to trade at the entrance of Hepu County People's Hospital. 'Er Ge' said that a person named "Twenty-six" took the heroin to the trading location. At about 14:30 on the same day, the defendant Zhu Youchuan (nicknamed 26) arrived at the entrance of Hepu County People's Hospital with heroin and called Chen to trade at the intersection of Dinghai Road and Pedestrian Street in Lianzhou Town, Hepu County, Guangxi Province. When the defendants Zhu Youchuan and Chen finished the trade at the above-mentioned location, they were arrested by the second brigade of the anti-drug detachment of the Beihai Municipal Public Security Bureau and the anti-drug brigade of the Yinhai Branch of the Beihai Municipal Public Security Bureau. They seized the two packets of heroin that the defendant Zhu Youchuan sold to Chen from Chen's body on the spot. The

suspected drug was weighed and weighed 0.51 grams. They also seized 200 yuan of drug money and 6 packets of suspected drug heroin from the defendant Zhu Youchuan. After weighing, the net weight was 0.38 grams, 0.42 grams, 0.08 grams, and one mobile phone that was used to conduct crimes. It was identified that the seized suspected drug detected heroin.

Part III

The defendant Zhu Youchuan also had no objection about the above facts during the trial, and the facts could also be proved by a household registration certificate, a certificate of return, a call record, a transcript, a list of seized items, a transcript of the weighing, a testimony of the witness Chen, and the statement of the defendant Zhu Youchuan as well as the confession, the physical and chemical inspection report and the notice of appraisal opinion, the on-site inspection transcripts and photographs, the identification of transcripts and other evidence. All the above things were enough to identify the facts.

Our court held that the defendant Zhu Youchuan violated the state's regulations on drugs and illegally sold heroin to others. His behaviour constituted a crime of drug trafficking. The public prosecutor accused the defendant Zhu Youchuan of committing drug trafficking. In the crime of joint drug trafficking, the defendant Zhu Youchuan played the main role and was the principal offender. He should be punished according to the law for all the crimes he participated in. Since the defendant Zhu Youchuan confessed his crimes truthfully after returning to the case, he could be punished softly according to law. The mobile phone that was used for drug trafficking should be confiscated according to law. Considering the facts, nature, situation, and damage to the society of the defendant's crime, in accordance with line items 1 and 4 of Article 347, line item 1 of Article 25, items 1 and 4 of Article 26, Article 52, Articles 53, line item 3 of Article 67 and Article 64 of the Criminal Law of the People's Republic of China, the sentences are as follows:

1. The defendant Zhu Youchuan committed the crime of drug trafficking and was sentenced to 10 months in prison and fined RMB 2,000. (The term of imprisonment shall be counted from the date of execution of the judgment. If the sentence is executed before January 6, 2019. The fine is limited to the law of this judgment. Payment within 10 days from the date of effectiveness. We may conduct mandatory payment if the pay is overdue).

2. One mobile phone transferred with the case will be confiscated and turned over to the state treasury.

If you disagree with this judgment, you may appeal to the Beihai Intermediate People's Court of Guangxi Zhuang Autonomous Region through the court or directly within 10 days from the second day after receiving the judgment. In the case of a written appeal, the original copy of the appeal shall be submitted in duplicate.

Judge Liu Weipei
6 June 2018
Judge Assistant Chen Haijuan
Clerk Huang Chuntian

C-2 Chinese Original

Part I

广西壮族自治区合浦县人民法院
刑事判决书

（2018）桂0521刑初277号

公诉机关广西壮族自治区合浦县人民检察院。

被告人朱有传，男，1984年7月12日出生于广西合浦县，汉族，初中文化，无业，住广西合浦县。因吸毒于2018年1月17日被北海市公安局银海分局处以行政拘留12日；于2018年1月29日被北海市公安局银海分局决定强制隔离戒毒二年。因涉嫌犯贩卖毒品罪于2018年3月7日被北海市公安局银海分局刑事拘留，同年3月19日被逮捕。现羁押于北海市第二看守所。

Part II

合浦县人民检察院以合检公刑诉［2018］236号起诉书指控被告人朱有传犯贩卖毒品罪，于2018年5月24日向本院提起公诉。本院依法适用简易程序，实行独任审判，公开开庭审理了本案。合浦县人民检察院指派检察员庞滨出庭支持公诉，被告人朱有传到庭参加诉讼。现已审理终结。

经审理查明：2018年1月16日14时左右，购毒人陈某电话联系绰号叫“二哥”的人要购买200元毒品海洛因，并约定在合浦县人民医院门口交易，“二哥”说其叫一名绰号“二十六”的人拿毒品海洛因到交易地点。当日14时30分左右，被告人朱有传(绰号二十六)持毒品海洛因到达合浦县人民医院门口，并打电话给陈某，约定在广西合浦县廉州镇定海路与步行街交汇处进行毒品交易，当被告人朱有传和陈某在上述地点交易完成后，被北海市公安局禁毒支队第二大队与北海市公安局银海分局禁毒大队民警现场抓获，当场从陈某身上缴获被告人朱有传贩卖给其的疑似毒品海洛因2小包，经称重，净重0.51克，并从被告人朱有传身上缴获毒资人民币200元和6小包疑似毒品海洛因，经称重，净重0.38克，0.42克，0.08克，作案手机一台。经鉴定，缴获的疑似毒品检出海洛因。

Part III

上述事实，被告人朱有传在开庭审理过程中也无异议，并有户籍证明，归案证明，通话记录，提取笔录，扣押物品清单，称量笔录，证人陈某的证言，被告人朱有传的供述，理化检验报告及鉴定意见通知书，现场勘验笔录及照片，辩认笔录等证据证实，足以认定。

本院认为，被告人朱有传违反国家对毒品的管理规定，非法向他人销售毒品海洛因，其行为构成贩卖毒品罪，公诉机关指控被告人朱有传犯贩卖毒品罪罪名成立。在共同贩卖毒品犯罪中，被告人朱有传起主要作用，是主犯，依法应按其所参与的全部犯罪处罚。被告人朱有传归案后如实供述自己的犯罪事实，依法可以从轻处罚。贩卖毒品使用的手机依法应当予以没收。根据被告人犯罪的事实、性质、情节和对社会的危害程度，依照《中华人民共和国刑法》第三百四十七条第一款和第四款、第二十五条第一款、第二十六条第一款和第四款、第五十二条、第五十三条、第六十七条第三款、第六十四条之规定，判决如下：

一、被告人朱有传犯贩卖毒品罪，判处有期徒刑十个月，并处罚金人民币2000元。

（刑期从判决执行之日起计算。判决执行以前先行羁押的，羁押一日折抵刑期一日，即自2018年3月7日起至2019年1月6日止；罚金限于本判决发生法律效力之日起十日内缴纳，逾期则强制缴纳）。

二、随案移送的手机一台，予以没收，上缴国库。

如不服本判决，可在接到判决书的第二日起十日内，通过本院或者直接向广西壮族自治区北海市中级人民法院提出上诉。书面上诉的，应提交上诉状正本一份，副本二份。

审 判 员　刘伟培

二〇一八年六月六日

法官助理　陈海娟

书 记 员　黄春添

D Judge Ethnicity Prediction

The judgments report judges' names but not their ethnicities. We train a model on census records containing names and ethnicities, then use it to predict whether judges belong to the Han majority or an ethnic minority.

D-1 Building the Classification Model

The 2005 census contains names and recorded ethnicities for nearly 13 million people. We set aside people whose surnames are predominantly Han, leaving about 2.5 million people whose surnames are shared by Han and minority groups. We randomly divide this sample into 90% for training and 10% for testing.

The model uses a long short-term memory (LSTM) neural network (Hochreiter and Schmidhuber, 1997). It reads each name character by character, learns patterns associated with ethnicity, and predicts the probability that the person belongs to an ethnic minority.[2]

Minorities make up only 7.2% of the training sample. We sample minority observations more often during training to help the model learn their names. We also use focal loss, which gives more weight to names that are difficult to classify (Lin et al., 2017).

The model's reported accuracy is 75% for the sample with shared surnames. Because Han names dominate this sample, we also assess how well the model distinguishes the two groups. Figure D1 shows the receiver operating characteristic (ROC) curve: the trade-off between correctly identifying minorities and incorrectly labeling Han individuals as minorities at different classification thresholds. The area under this curve (AUC) is 0.77 in the test set; 0.5 indicates random guessing and 1 indicates perfect classification.

D-2 Predicting Judges' Ethnicities

We apply the classification procedure to judges' names and calculate the predicted Han share in each province. Figure D2 compares this share with the Han population share in the 2005 census. Provinces with larger Han populations also have larger predicted shares of Han judges ($R^2 = 0.898$).

The predictions reproduce broad geographic patterns, but this does not verify individual judges' ethnicities. Their actual ethnicities are unobserved, and judges may differ in ethnic composition from the local population.

[2]The network uses character embeddings, 10% dropout, an LSTM layer, two linear layers separated by a ReLU activation, and a final softmax layer. We train it for 3,500 iterations with batches of 64 names.

Figure D1: Classification Model Performance: ROC Curve

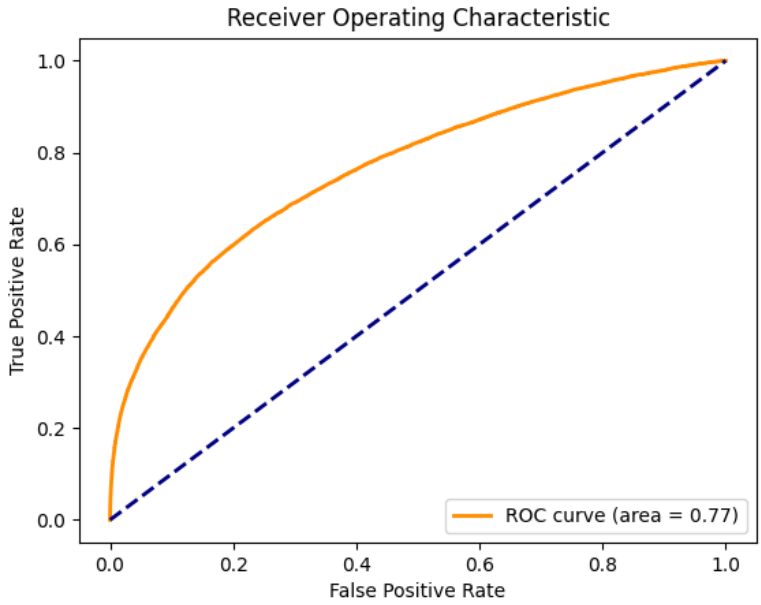
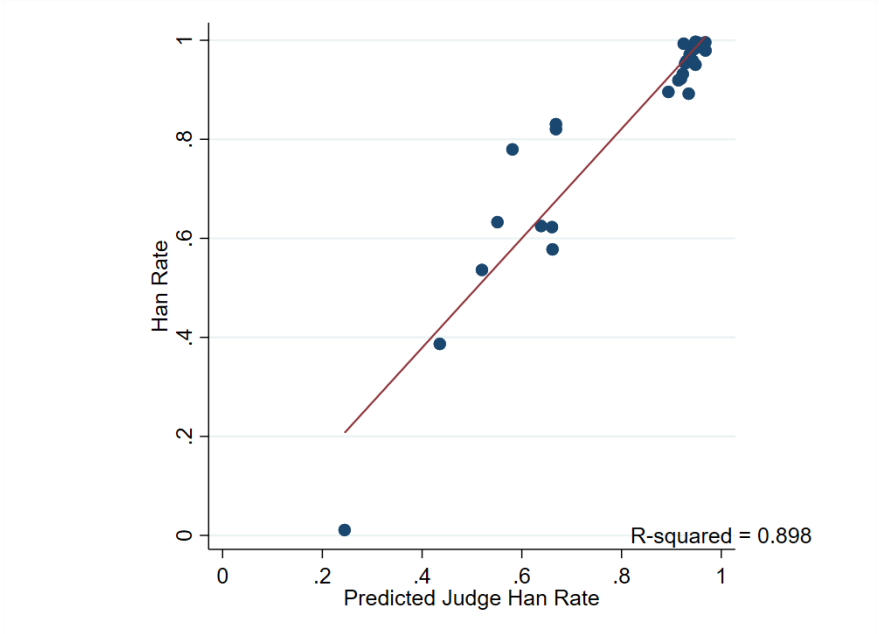
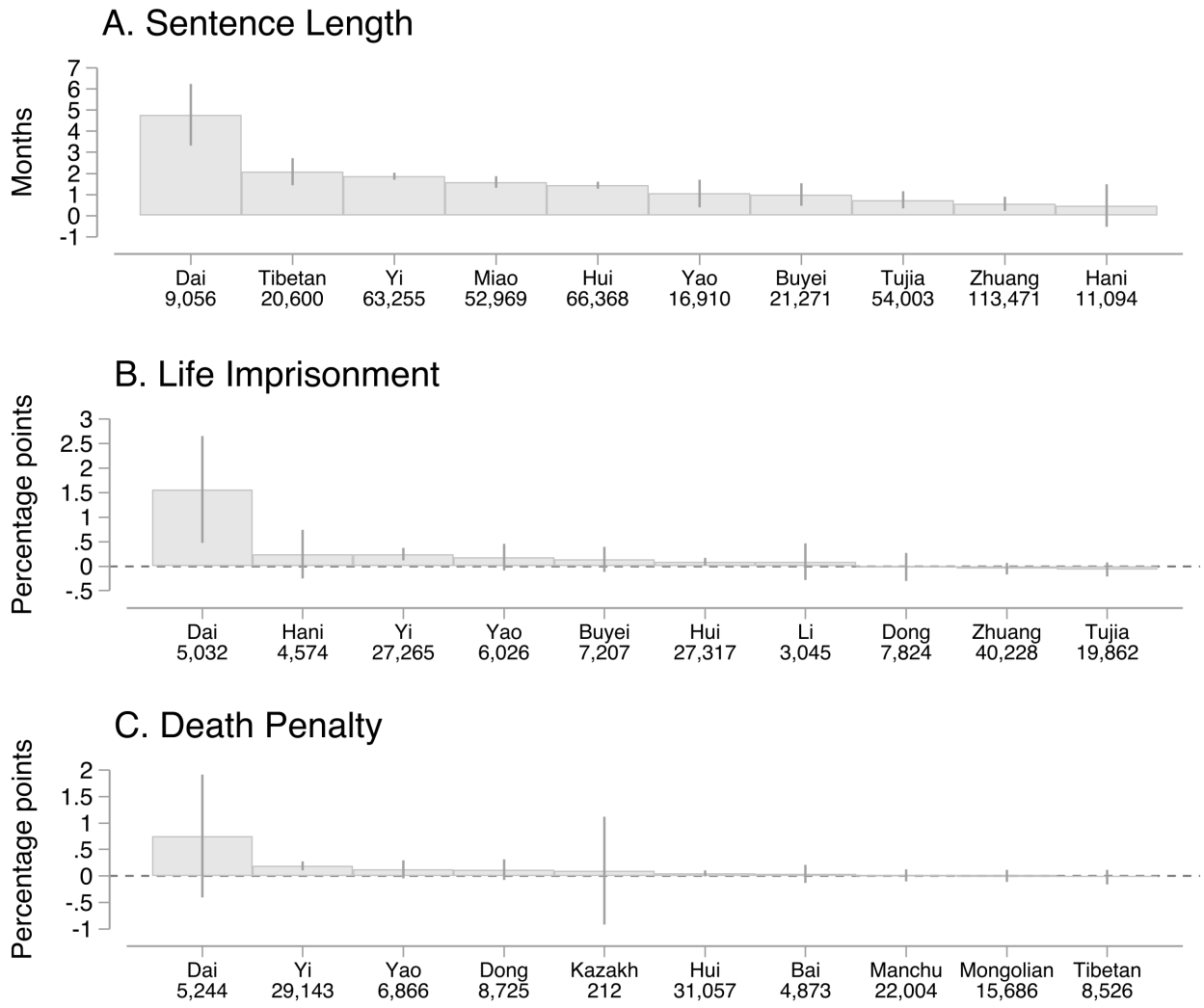


Figure D2: Predicted Han Share among Judges and Han Population Share



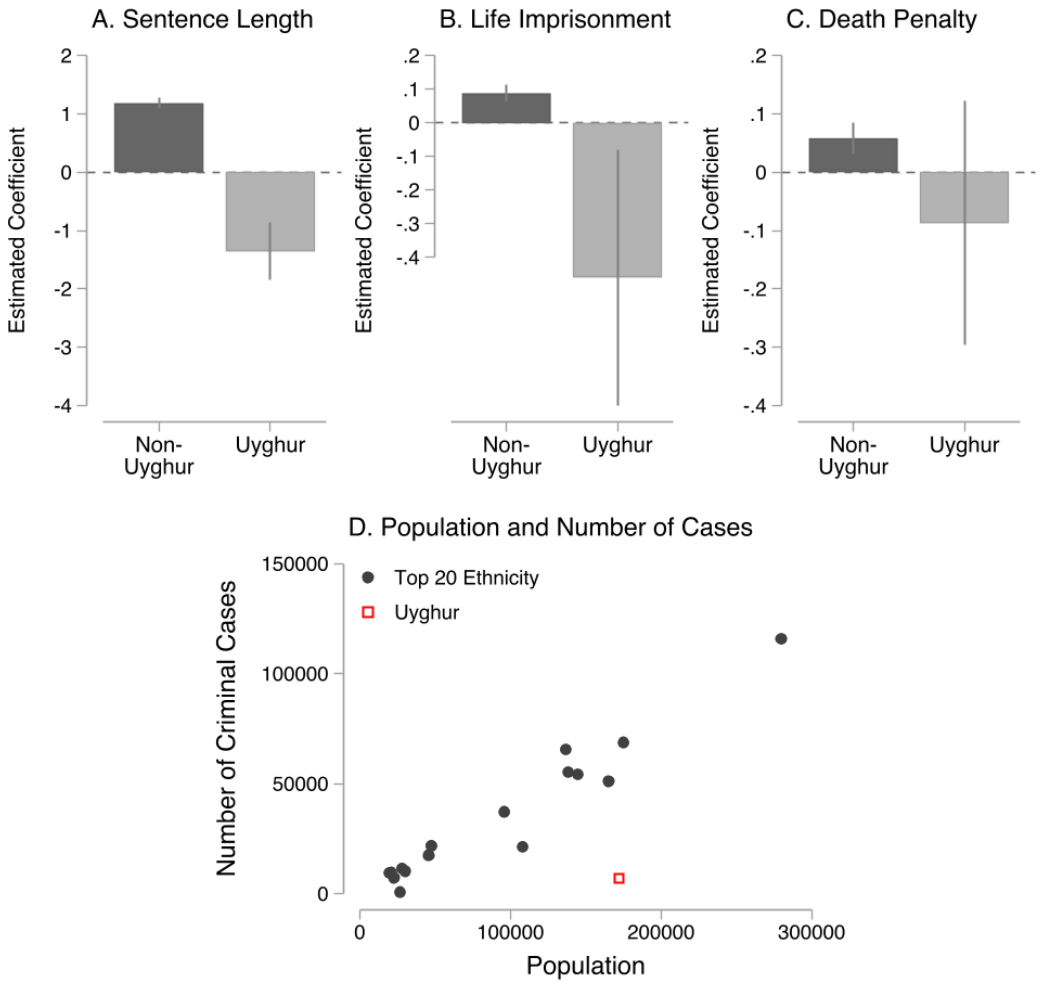
E Additional Results on Sentencing Disparities

Figure E1: Minority Penalties among the 20 Largest Ethnic Groups



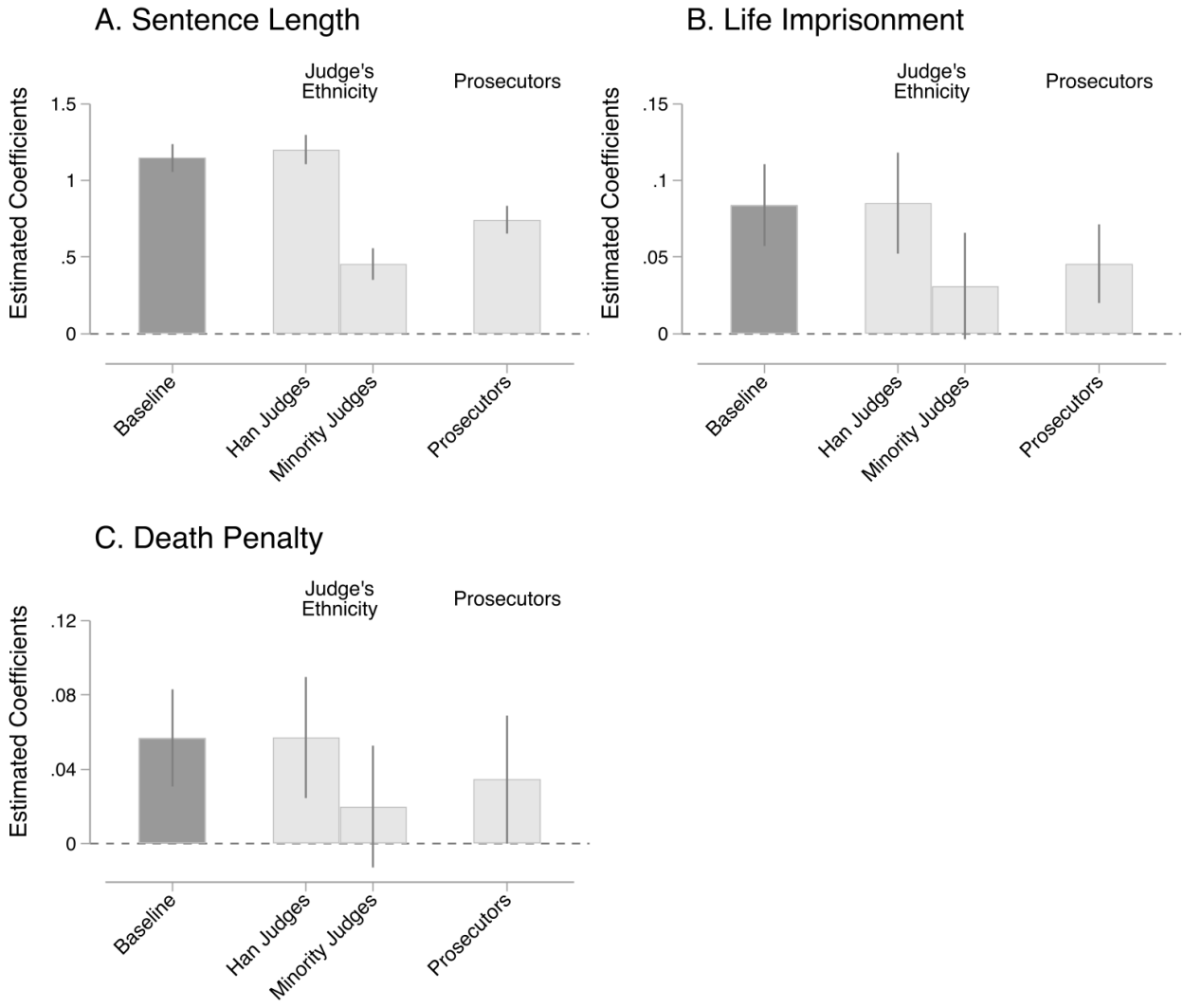
Notes: Eligible groups are the 20 largest ethnic minorities by national population in the 2010 census (NBSC, 2012, Table 1-13). Each panel displays the ten largest point estimates among these groups, without a significance cutoff. Each regression compares one minority group with Han defendants using equation 1, with the same controls and fixed effects. Panel A reports months; Panels B and C report percentage points and restrict the samples to eligible offenses. Numbers below the group names count minority defendants in each estimation sample. Bars show coefficients; whiskers show pointwise 95% confidence intervals with standard errors clustered by court.

Figure E2: Facts: Sentencing Disparity for Uyghur Ethnicity



Notes: This figure presents the additional penalties imposed on Uyghur and non-Uyghur ethnic minorities compared to Han defendants. For each group, we estimate equation 1 using only cases involving Uyghur or non-Uyghur minority defendants and the Han majority. The outcome variables are sentence length in months (Panel A), life imprisonment (Panel B), and the death penalty (Panel C). For life imprisonment and the death penalty, we restrict the samples to crime types eligible for these sentencing penalties. The bars represent the estimated coefficients, while the lines show the 95% confidence intervals. Standard errors are clustered at the court level. Panel D reports cases per population, plotting the number of criminal cases in our baseline data (Y-axis) against the population (X-axis) for each minority group.

Figure E3: Judge's Ethnicity, Prosecutors, and Minority Penalty



Notes: This figure presents key findings on the facts about ethnic minority issues in China's criminal justice system. The outcome variables are sentence length in months (Panel A), life imprisonment (Panel B), or the death penalty (Panel C). For life imprisonment and the death penalty, we limit the samples to crime types eligible for these sentencing penalties. The bars represent the estimated coefficients, while the lines show the 95% confidence intervals. Standard errors are clustered at the court level. For the judge's ethnicity, we estimate the baseline regression using each subsample. For prosecutors, we use the recommended sentences as the outcome variables. See the main text for additional details. Panels E and F of Table E1 report the estimates and standard errors.

Table E1: Facts about Ethnic Minority Issues in China's Criminal Justice System

	Sentence Length		Life Imprisonment		Death Penalty	
	Coef. (s.e.)	% of Han mean	Coef. (s.e.)	% of Han mean	Coef. (s.e.)	% of Han mean
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. All						
All Crimes	1.147***	6.8%	0.084***	6.5%	0.057***	14.0%
	(0.046)		(0.014)		(0.013)	
Panel B. Crime Types						
Violent Crime	1.462***	7.0%	0.101***	7.8%	0.072***	17.7%
	(0.068)		(0.025)		(0.022)	
Drug Crime	1.604***	5.3%	0.145***	11.2%	0.078***	19.2%
	(0.087)		(0.037)		(0.030)	
Sex Crime	1.406***	6.0%	0.084***	6.5%	0.038**	9.4%
	(0.096)		(0.023)		(0.019)	
Property Crime	1.113***	8.2%	0.036**	2.8%	0.024	5.8%
	(0.049)		(0.016)		(0.017)	
Others	0.342***	2.4%	0.022	1.7%	0.010	2.4%
	(0.050)		(0.020)		(0.015)	
Panel C. Ethnic Groups						
Christianity	1.552***	9.2%	0.111***	8.6%	0.075***	18.5%
	(0.056)		(0.030)		(0.024)	
Islam	1.255***	7.4%	0.080**	6.2%	0.067**	16.6%
	(0.067)		(0.036)		(0.027)	
Other Religions	1.000***	5.9%	0.032	2.5%	0.024	5.8%
	(0.047)		(0.024)		(0.021)	
Conflict History	1.482***	8.8%	0.120***	9.2%	0.101***	25.0%
	(0.057)		(0.029)		(0.021)	
Non-conflict History	0.939***	5.6%	0.047**	3.6%	0.028	6.9%
	(0.050)		(0.020)		(0.019)	

Panels D–F and notes continue on the next page.

Table E1 (continued)

	Sentence Length		Life Imprisonment		Death Penalty	
	Coef. (s.e.)	% of Han mean	Coef. (s.e.)	% of Han mean	Coef. (s.e.)	% of Han mean
	(1)	(2)	(3)	(4)	(5)	(6)
Panel D. Locations						
High Income	0.519***	3.2%	0.042**	3.2%	0.023	5.6%
	(0.0538)		(0.018)		(0.018)	
Low Income	1.352***	7.7%	0.094***	7.3%	0.084***	20.8%
	(0.0511)		(0.023)		(0.021)	
High Minority Share	1.406***	7.9%	0.088***	6.8%	0.077***	19.0%
	(0.0480)		(0.015)		(0.016)	
Low Minority Share	0.598***	3.7%	0.046	3.5%	0.029	7.2%
	(0.0502)		(0.033)		(0.020)	
High Crime Rate	1.307***	7.9%	0.090***	6.9%	0.068***	16.8%
	(0.0467)		(0.019)		(0.017)	
Low Crime Rate	0.727***	4.2%	0.049**	3.8%	0.042**	10.4%
	(0.0485)		(0.021)		(0.017)	
Panel E. Ethnicity of Judges						
Han Judge	1.201***	7.1%	0.085***	6.5%	0.057***	14.1%
	(0.048)		(0.017)		(0.017)	
Minority Judge	0.453***	2.8%	0.031*	2.4%	0.02	4.9%
	(0.052)		(0.018)		(0.017)	
Panel F. Prosecutor's Recommended Sentences						
Prosecutors	0.741***	3.0%	0.046***	3.5%	0.035**	8.5%
	(0.0464)		(0.013)		(0.018)	

Notes: This table reports estimates of equation (1). Sentence length is measured in months; life-imprisonment and death-penalty coefficients and standard errors are in percentage points. Sentence-length percentages use the Han mean in each estimation sample. For life imprisonment and the death penalty, all rows use the Han rates in the respective all-crime baseline estimation samples: 1.300% and 0.405%. Court-clustered standard errors appear in parentheses. Life-imprisonment and death-penalty samples include only eligible offenses. Panel B estimates separate regressions by crime type. Panel C compares defendants from the minority groups in each religion or conflict-history category with all Han defendants. Panel D splits prefectures at the median income, minority population share, or per capita case count. Panel E splits the sample by judge ethnicity; Panel F uses prosecutors' recommended sentences as outcomes. All specifications include crime-category, year-month, province, court, and judge fixed effects and case and defendant controls. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

E-1 Baseline Estimates and Successive Controls

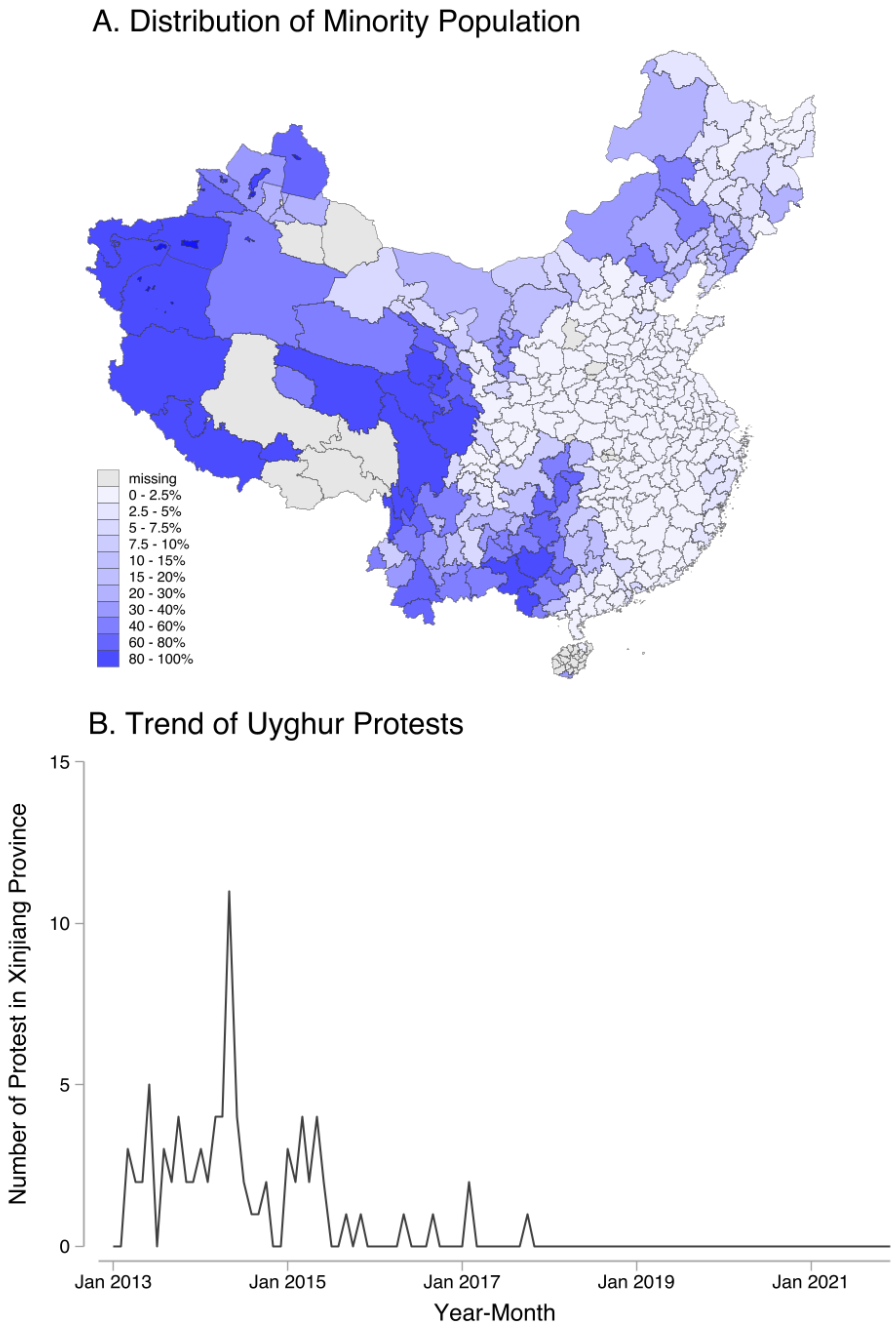
Table E2: Facts—What Contributes to Unconditional and Conditional Ethnic Disparity?

	Dependent variable: Sentence Outcomes					
	(1)	(2)	(3)	(4)	(5)	(6)
A. Effects on sentence length in month						
minority	2.902***	0.711**	0.818***	0.676***	0.850***	1.147***
	(0.412)	(0.292)	(0.275)	(0.255)	(0.062)	(0.046)
% of Han mean	17.2%	4.2%	4.8%	4.0%	5.0%	6.8%
obs	6,440,961	6,439,124	6,439,124	6,439,124	6,439,118	6,434,145
B. Effects on life imprisonment (percentage points)						
minority	0.214	0.257**	0.242**	0.057	-0.062*	0.084***
	(0.139)	(0.126)	(0.113)	(0.088)	(0.034)	(0.014)
% of Han mean	16.4%	19.7%	18.6%	4.4%	-4.8%	6.5%
obs	1,451,469	1,450,699	1,450,699	1,450,699	1,450,680	1,446,093
C. Effects on death penalty (percentage points)						
minority	0.289***	0.190**	0.189**	0.108**	0.065***	0.057***
	(0.086)	(0.077)	(0.074)	(0.053)	(0.017)	(0.013)
% of Han mean	71.4%	47.0%	46.8%	26.6%	16.1%	14.0%
obs	1,963,066	1,962,011	1,962,011	1,962,011	1,961,995	1,957,249
Crime category FE	no	yes	yes	yes	yes	yes
Case characteristics	no	yes	yes	yes	yes	yes
Defendant characteristics	no	no	yes	yes	yes	yes
Year-month FE	no	no	yes	yes	yes	yes
Province FE	no	no	no	yes	yes	yes
Court FE	no	no	no	no	yes	yes
Judge FE	no	no	no	no	no	yes

Notes: This table reports the baseline results estimating equation (1). The outcome variables are sentence length in months (Panel A), life imprisonment (Panel B), and the death penalty (Panel C). For life imprisonment and the death penalty, we restrict the samples to eligible crime types and report coefficients and standard errors in percentage points. Relative gaps divide the coefficient by the Han outcome mean in the same estimation sample, using consistent units. We gradually add control variables and fixed effects from Column (1) to Column (6). Column (1) shows the unconditional gap, and Column (6) represents our baseline conditional gap presented in Table E1 and Figure 2. Standard errors are clustered at the court level. * significant at 10%; ** significant at 5%; *** significant at 1%.

F Supplementary Material for the Uyghur-Unrest Analysis

Figure F1: Variations in DDD: Minority Share and Uyghur Protests



This figure illustrates the variations used in the Uyghur unrest quasi-experiment design. Panel A shows the geographical distribution of minority population shares across prefectures. Panel B presents the time trend of Uyghur unrest in Xinjiang Province, China, from 2013 to 2021.